

Hypothesis Testing as Bivariate Analyses

j3
– 30.09.26 –



Plateforme OMICS - MIO

Bio-informatique & Sciences de l'Environnement : Exploration de la Diversité Taxonomique des Ecosystèmes par Metabarcoding



Variability in species richness between the 2 stations, depths



Is there a **real significant** difference or
just a coincidence ?



Using **statistics** to answer your question !!

Population VS samples

Population: set of individuals or objects of the same kind (very large or infinite)

- We can't study an entire population: in statistics, we study a limited number of individuals, a part of the population: **a sample**
- We try to **deduce properties** of the population from the sample
- If we want to **study the variability** of a variable of interest in the population, we need a **representative sample** (drawn at random)

In a population, we can measure a characteristic: **a variable** that is the result of a random phenomenon.

- Qualitative
- Quantitative (continuous)

A **probability law** describes the random behavior of a phenomenon that depends on chance.

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- Qualitative
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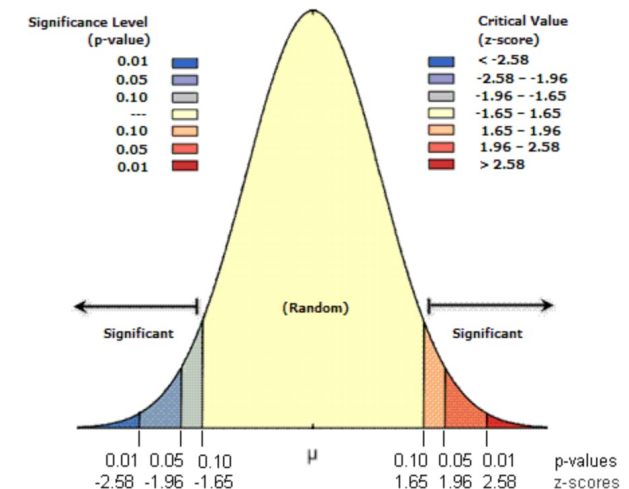
A **probability law** describes the random behavior of a phenomenon that depends on chance.

THE NORMAL LAW

If we have 1000 samples of a variable following a normal distribution, and plot the number of samples equal to each value, we obtain a "bell" curve / gaussian distribution

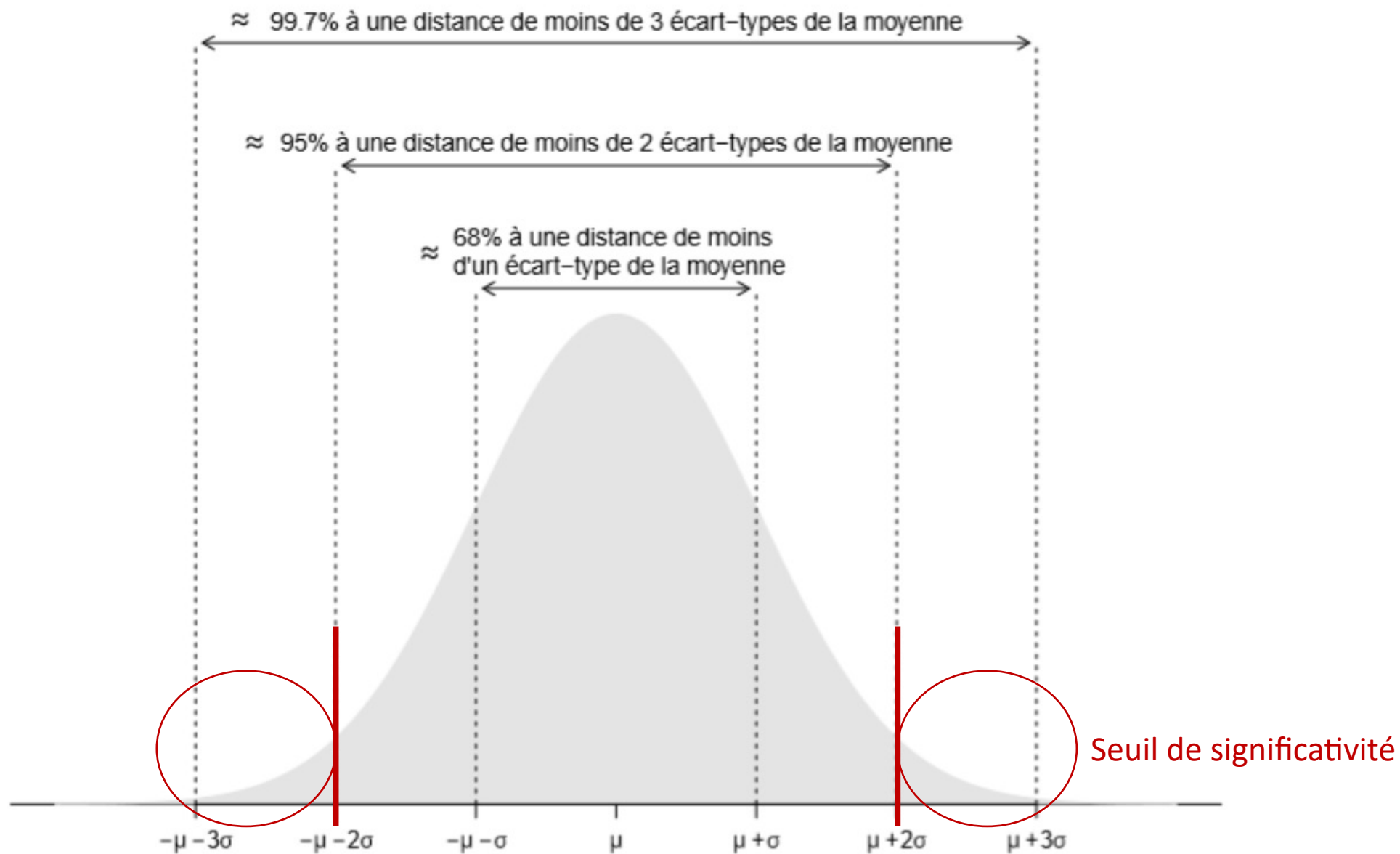
$X \sim N(\mu, \sigma^2)$ with μ and σ^2 the parameters of the distribution:

- μ : expectation of X
- σ = dispersion around the mean = Standard deviation (SD) of X



- Symetric, unimodal
- Center around the mean

Répartition des valeurs autour de la moyenne

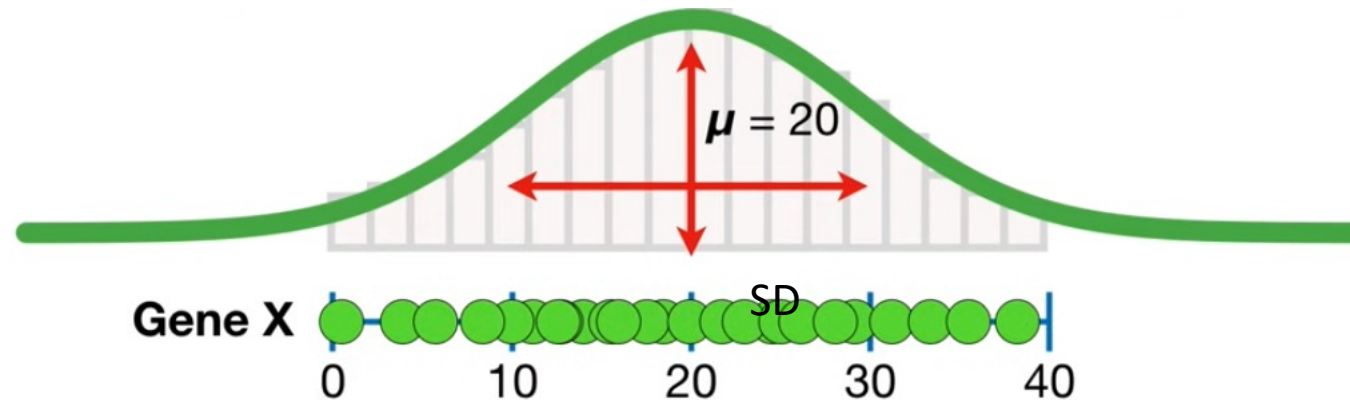


Remember : Descriptive statistics (Univariate analysis)



Merely describe, show and summarize collected data

- **Central tendency** (mean, median...)
- **Dispersion** (variance, standard deviation)
- **Frequency distribution** (count, relative, cumulative)

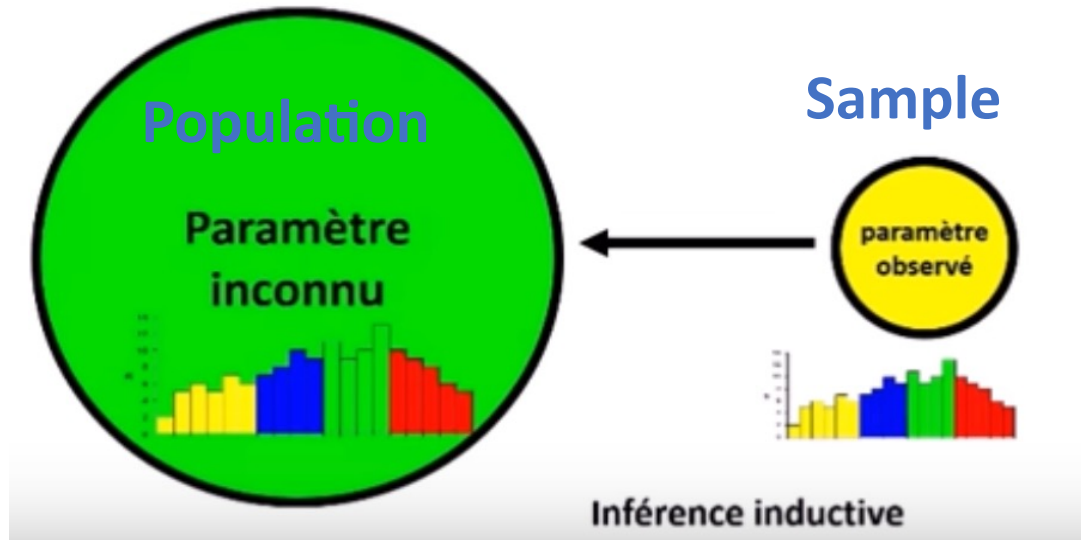
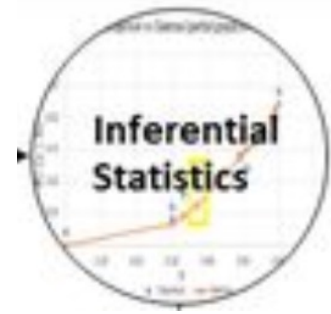


Identify the characteristics of data for each variable(s)

→ Allows you to formulate hypotheses and guide statistical analyzes

Inferential Statistics

Predictions - Generalizations



Make inferences about the population

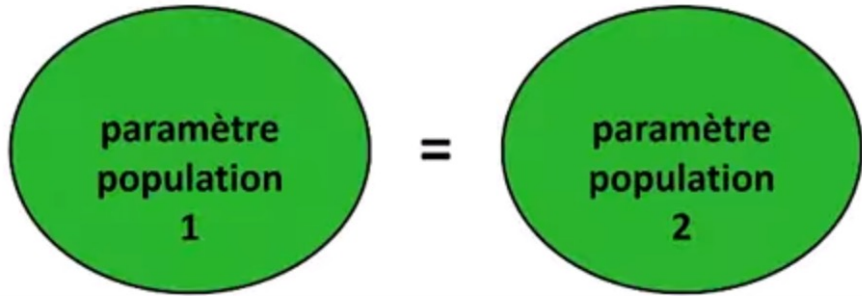
- How can I use my sample to make predictions about the population = **Estimation**
- How do I prove a theory about my data's behaviour (comparison) = **Hypothesis Testing**

Hypothesis testing approach

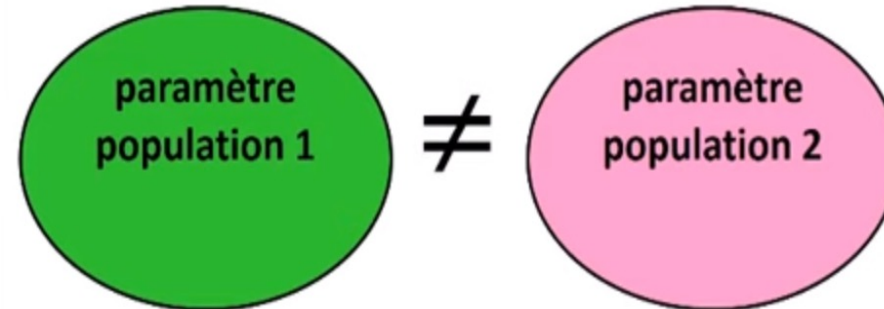
Trying to validate a hypothesis relating to a population parameter from a sample comparisons

Is there a **real** difference or just a coincidence (chance)

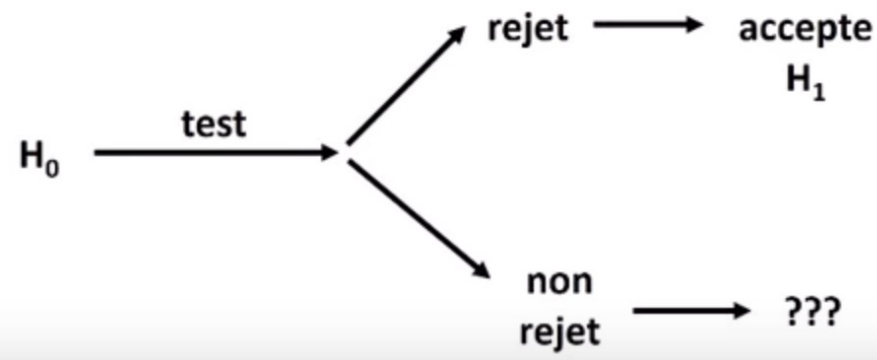
Null Hypothesis H_0



Alternative hypothesis H_1



We are testing the null hypothesis!

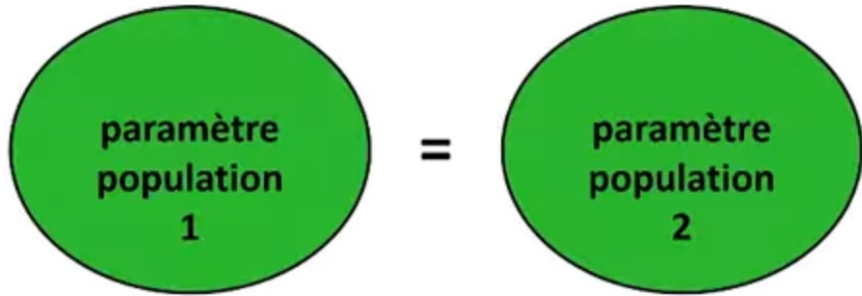


Hypothesis testing approach

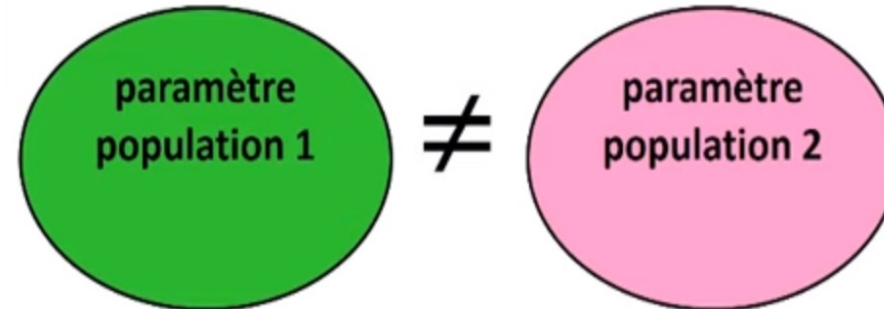
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Null Hypothesis H0



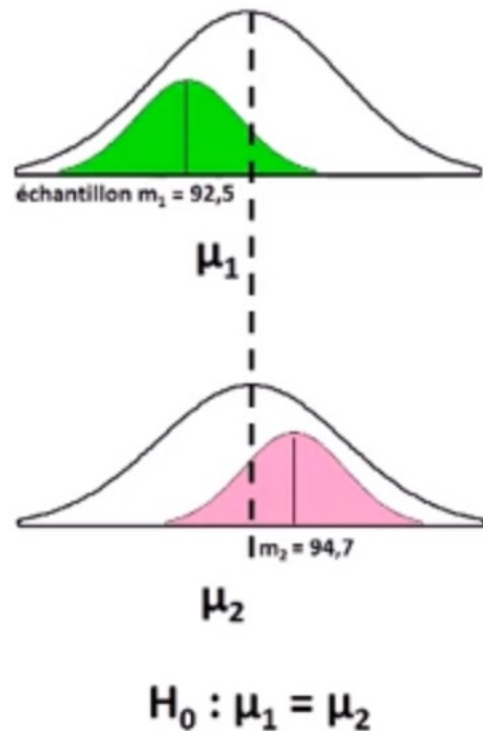
Alternative hypothesis H1



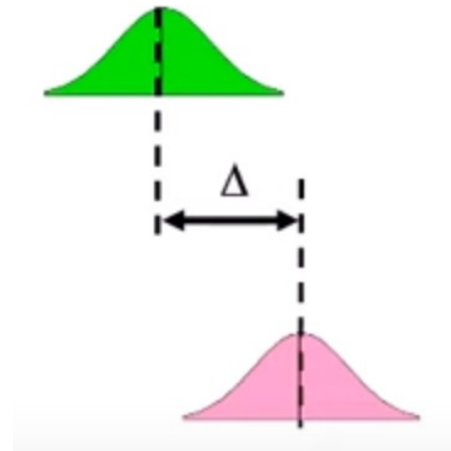
“Absence of Evidence is not Evidence of Absence”

Hypothesis testing & mean comparison

If H_0 true... no difference

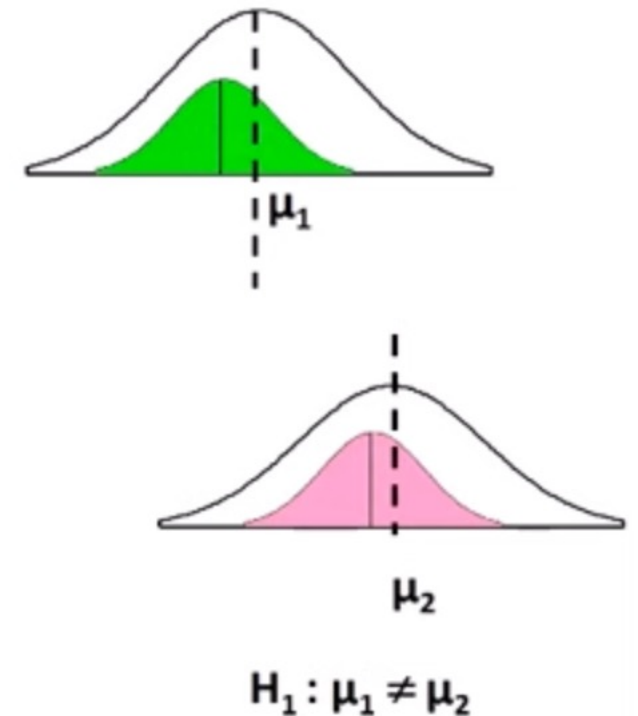


SAME distribution
→ **Sampling fluctuation**



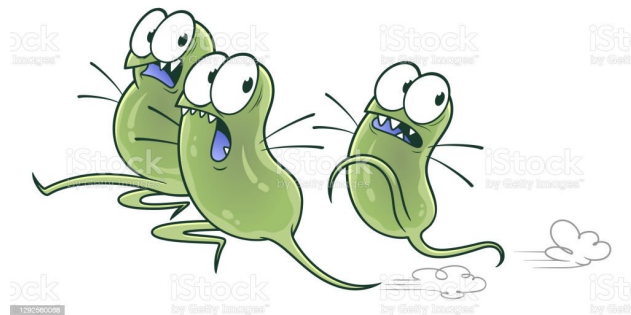
Si H_0 vraie : $\Delta = m_1 - m_2 \approx 0$

If H_0 rejected, H_1 accepted



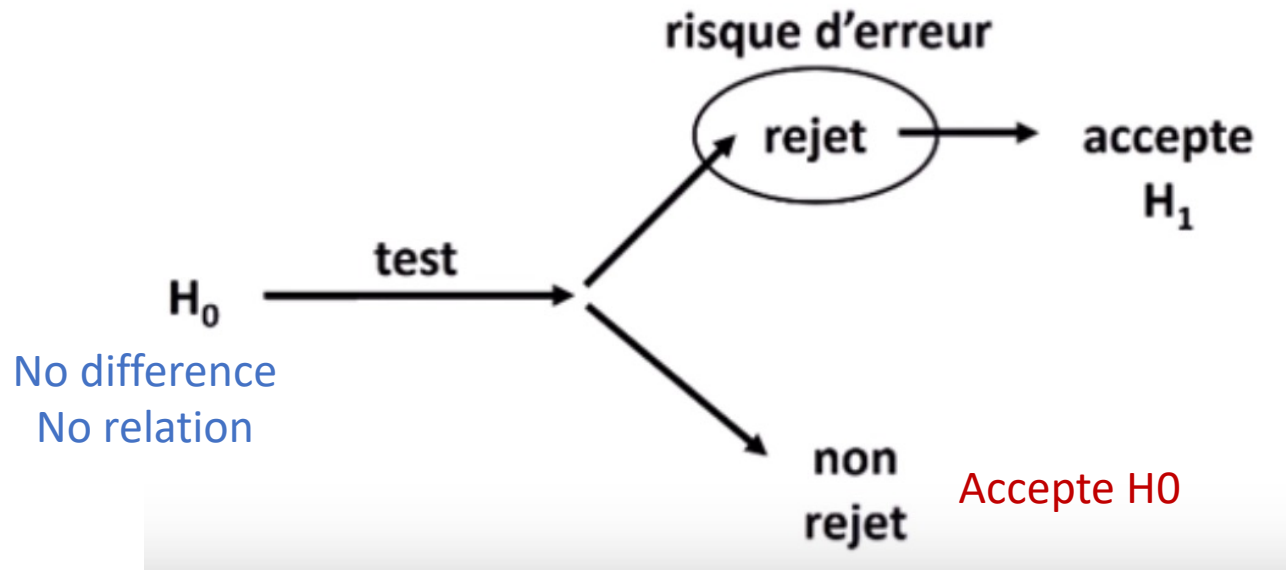
Two different distributions

Inference Issue : Subjected to errors!!
The risk is linked to the result of hypothesis testing
Because of your sampling!



The risk of Type I error α

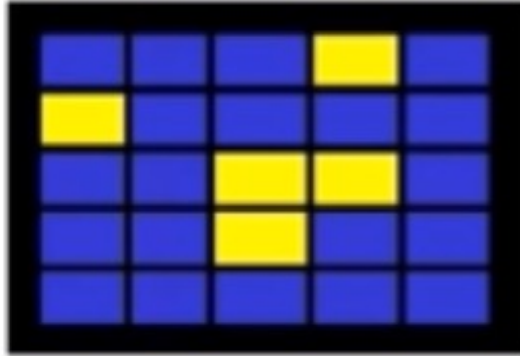
- A probability between 0 and 1, or 0 and 100%
- Is when a difference is affirmed but there is none (=False positive)!!



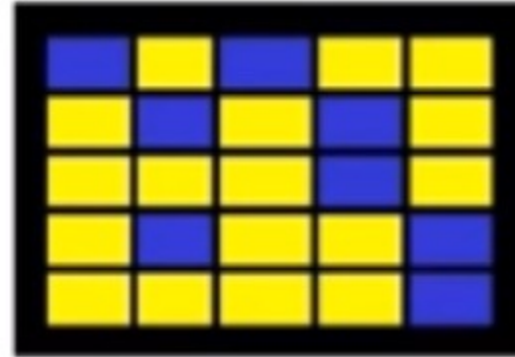
α = Risk to reject H_0 if H_0 is true

Sampling

25 tiles
→ 80% blue



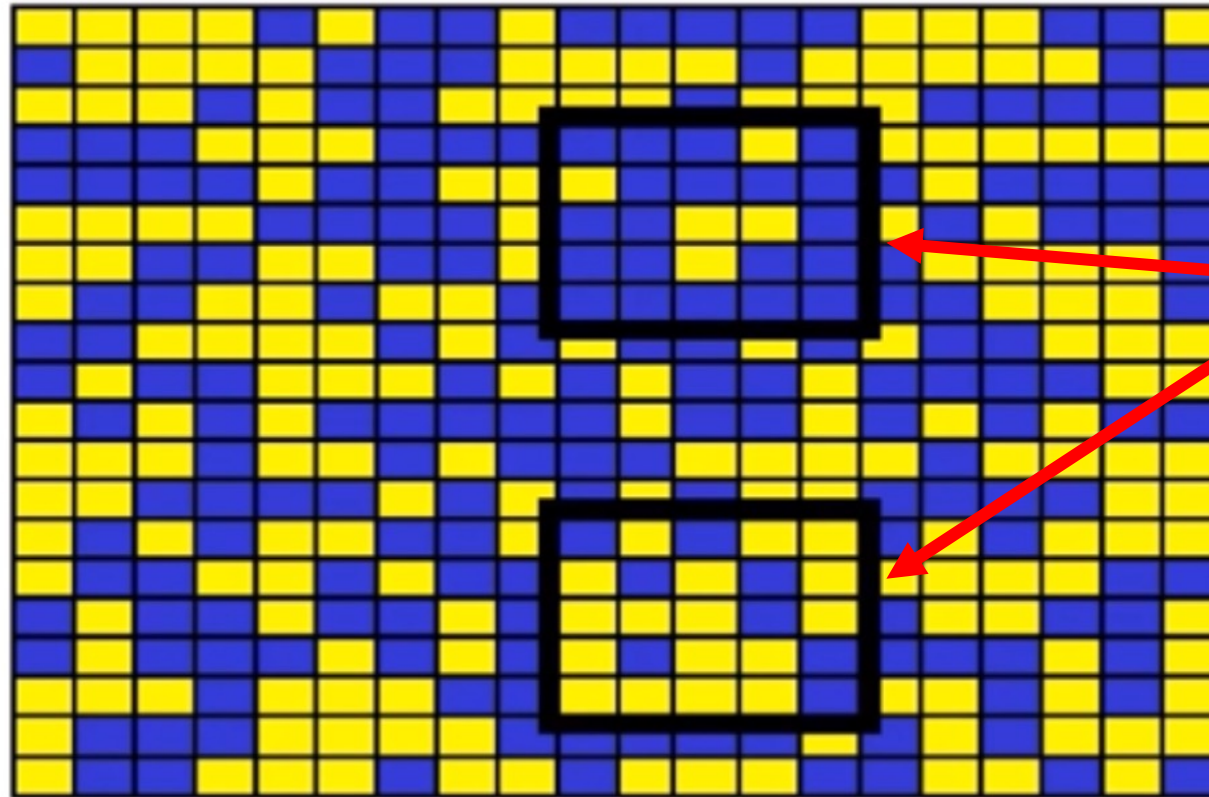
25 tiles
→ 32% blue



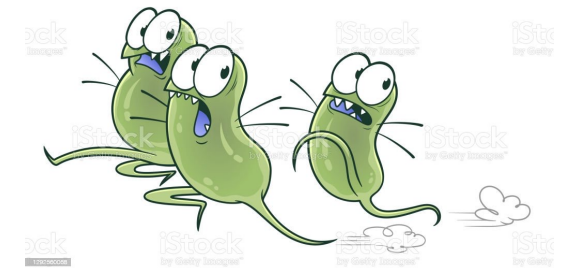
Do the two samples come from the same population? (same distribution)?

- **H0 is rejected**
- but let's go to the store...see the population

Come from the same population (50% blue, 50 % yellow)!!

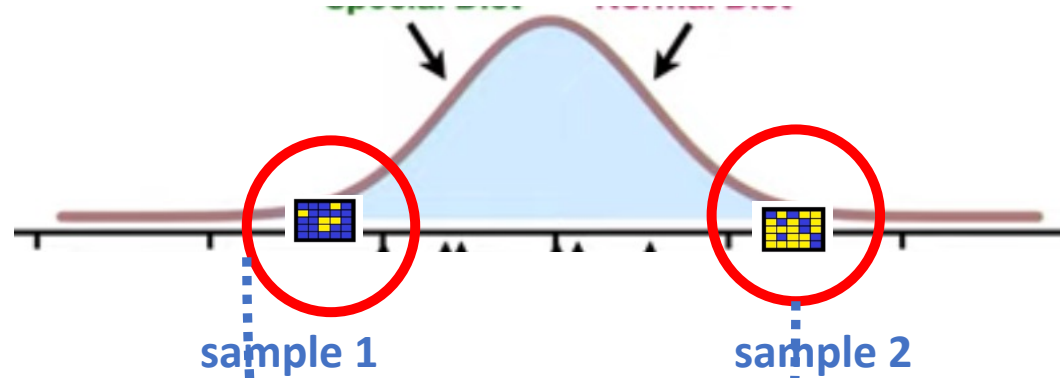


Rare sample type

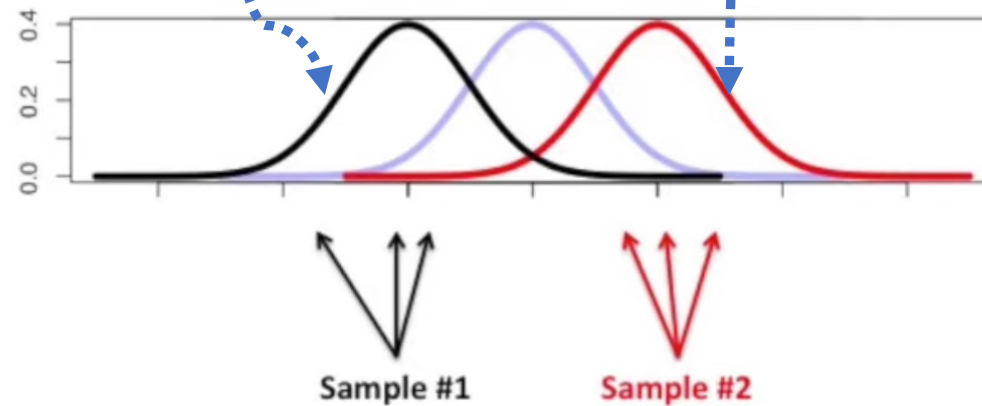


Conclude on the basis of our samples that they came from two different distributions
= Type I error

Data come from the same distribution but ...

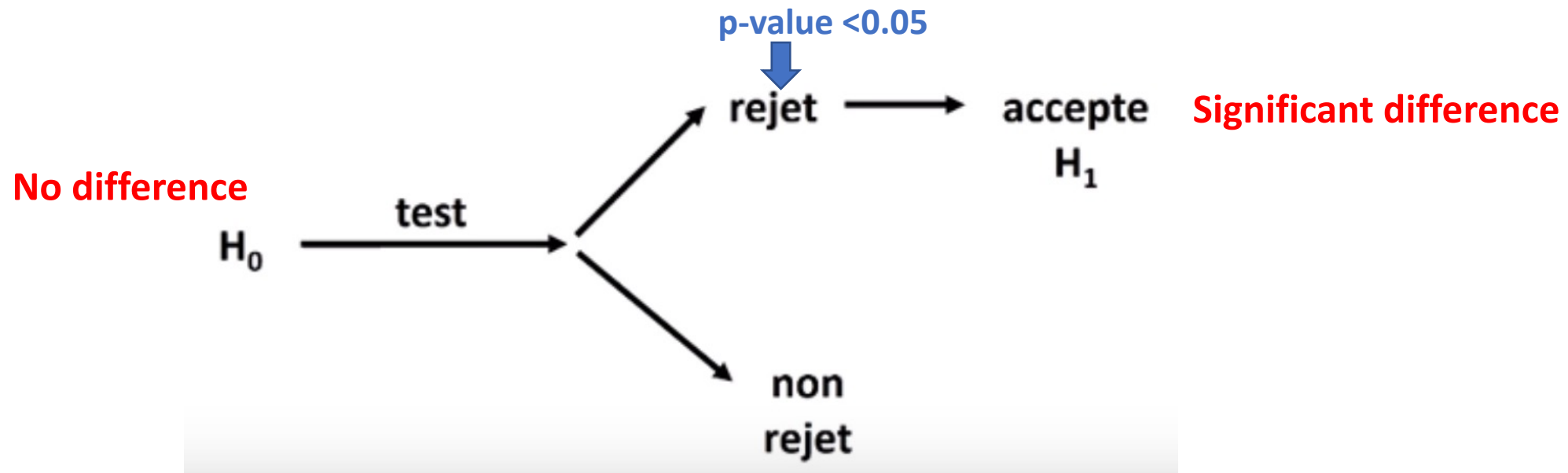


The test see...



Two different distributions

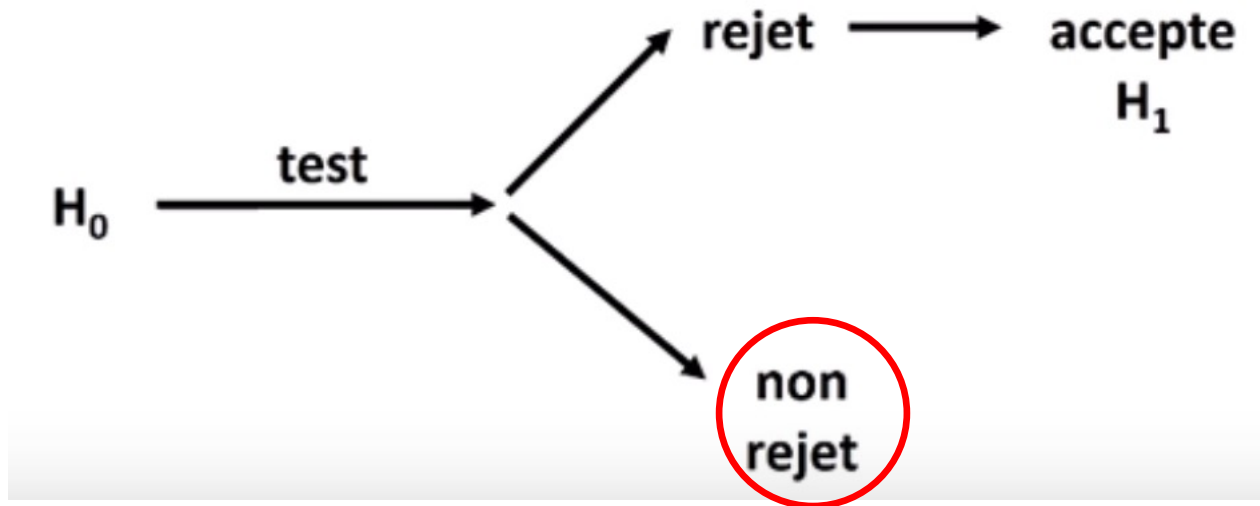
- α is chosen before the test : **Significance threshold**
- α often set 5% (H_0 wrongly rejected)
- In science the "almost no chance" translates to in less than 5% of cases where H_0 is true = **p-value < 0.05**



Risk of Type II Error : β

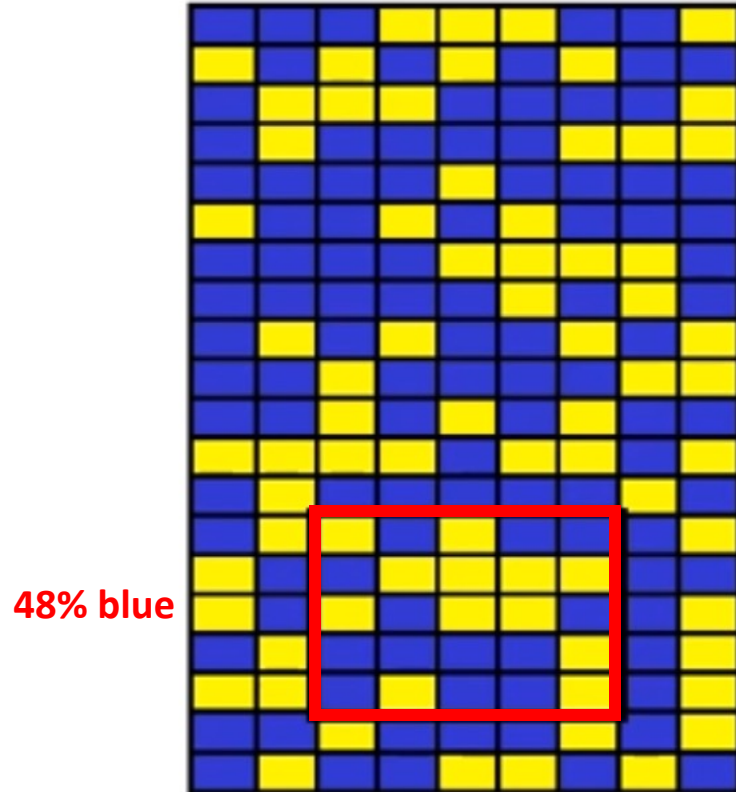
Failing to conclude a difference when there is a true one ("False Negative")

Probability of not rejecting H_0 , if H_1 is true



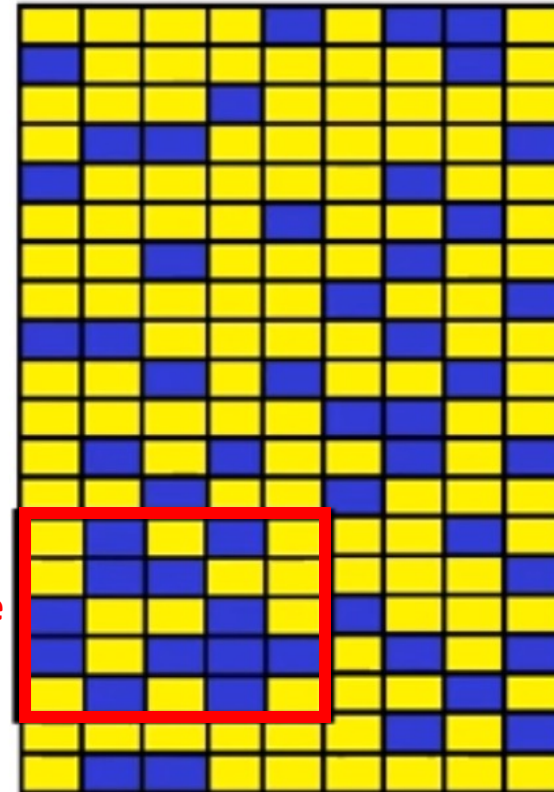
β is not calculable

60% of blue



48% blue

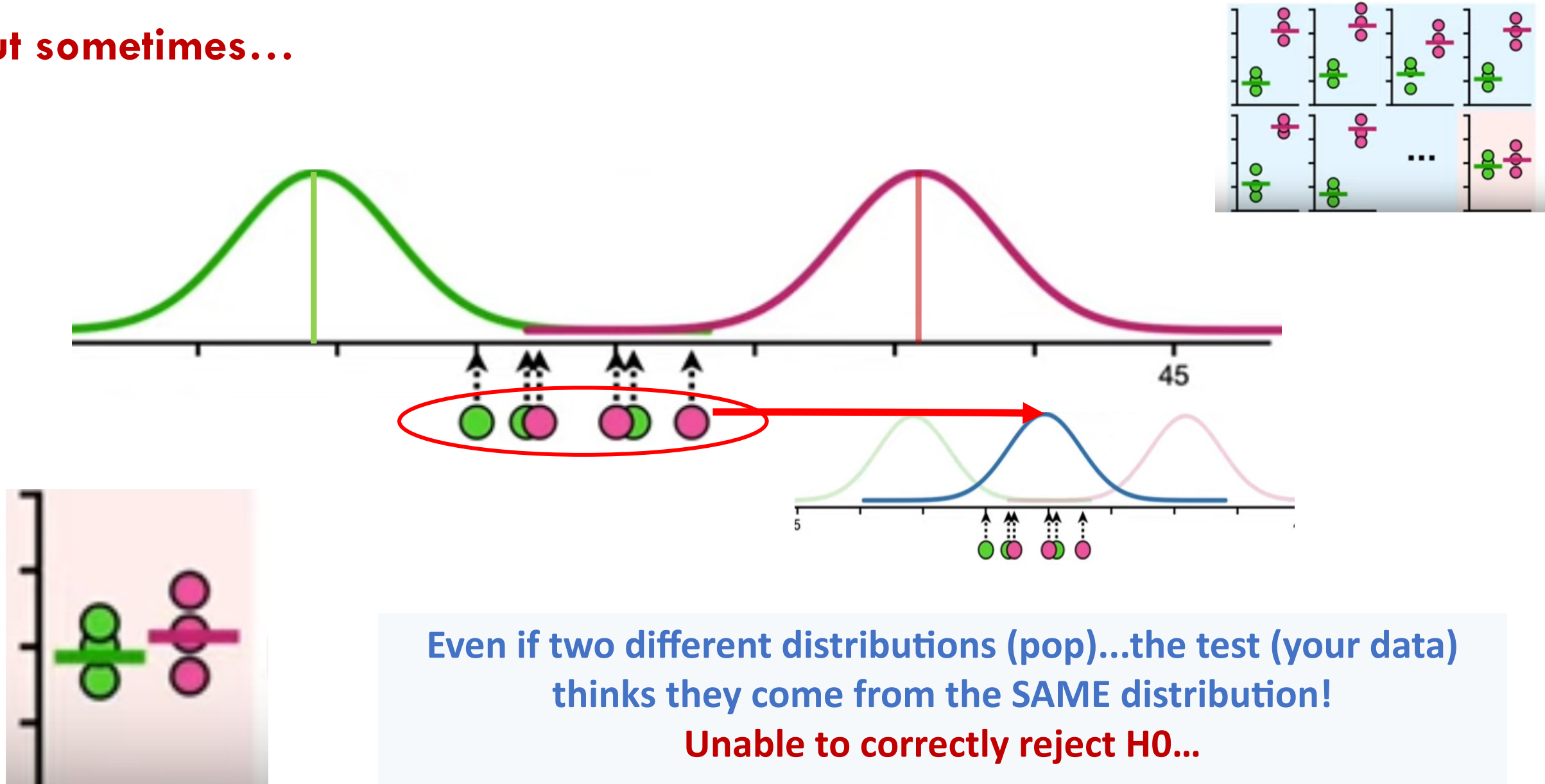
30% of blue



52% blue

- 2 different tiles = 2 different populations, H_0 should be rejected But that would not have been the case during the test with our sampling...

But sometimes...

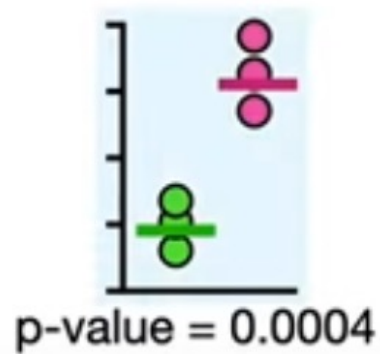
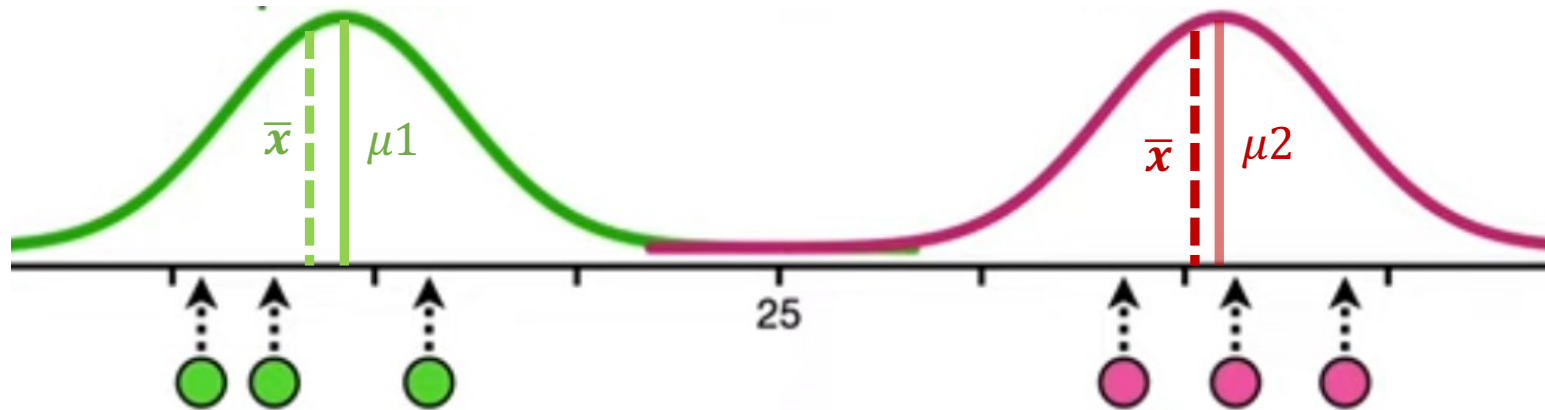


$p=0.23!!!$

Even if two different distributions (pop)...the test (your data) thinks they come from the SAME distribution!

Unable to correctly reject H_0 ...

Scientifically ... representative sampling of population



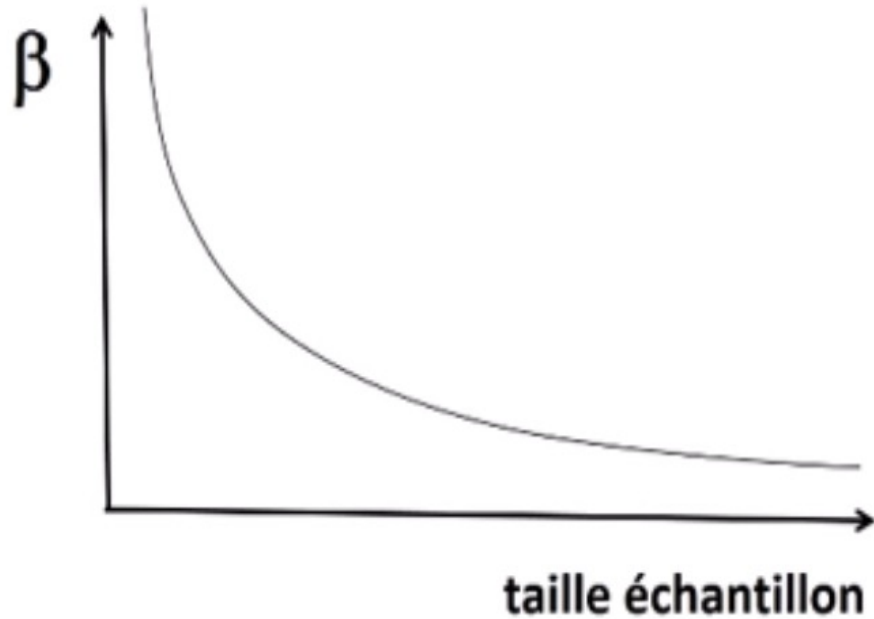
→ H_0 correctly rejected

→ = Data do not belong to same distribution

→ **Two different populations**

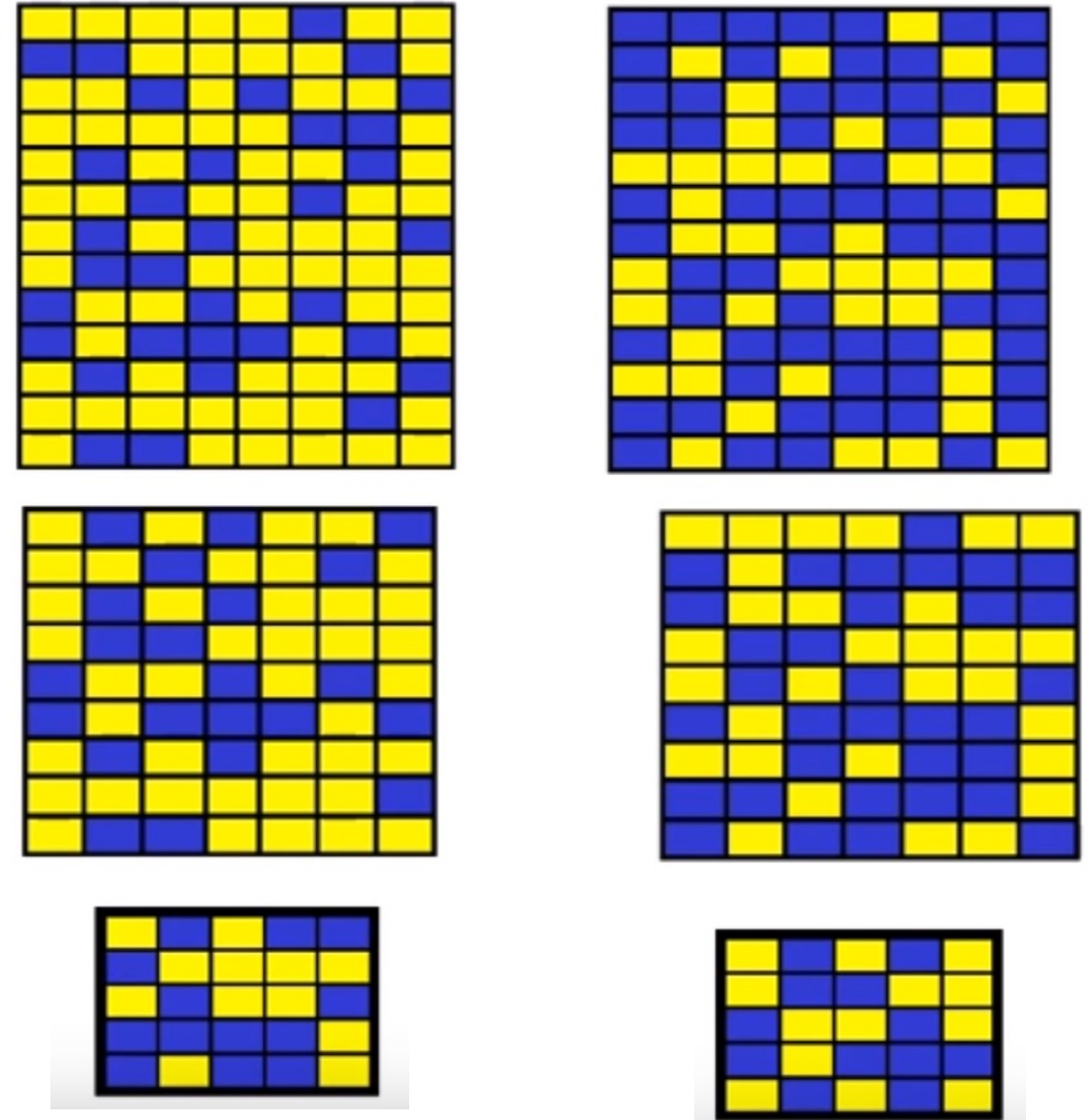
Fundamental relationship

$$\text{Power} = 1 - \beta$$



Power: Probability of correctly reject the H_0 hypothesis
Ability of a test to detect differences

The more the size increases, the more the differences appear! The power of the test increases!

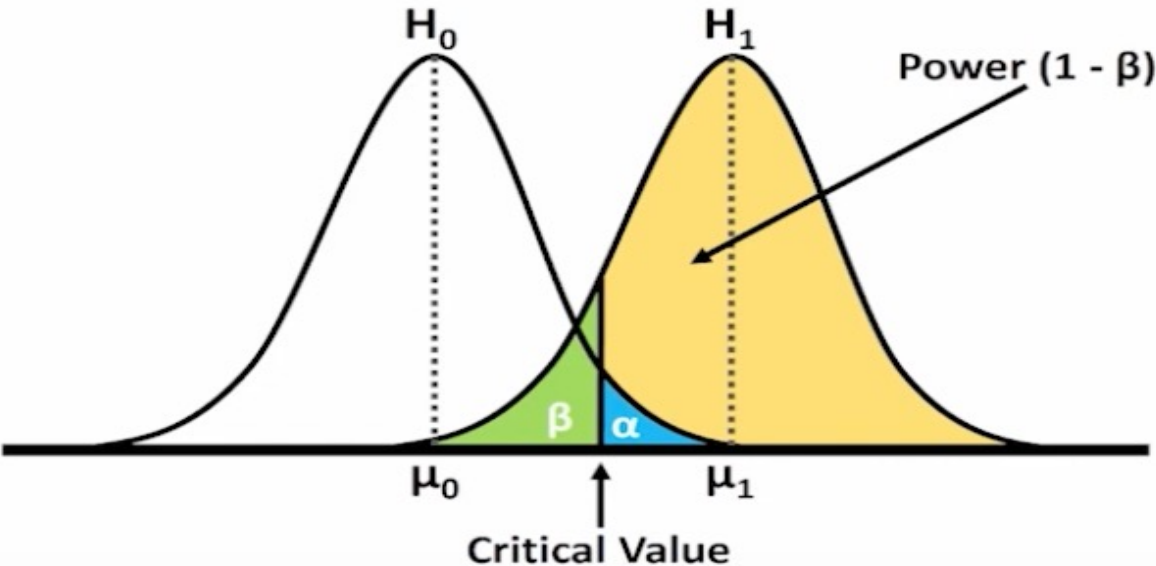


Summary

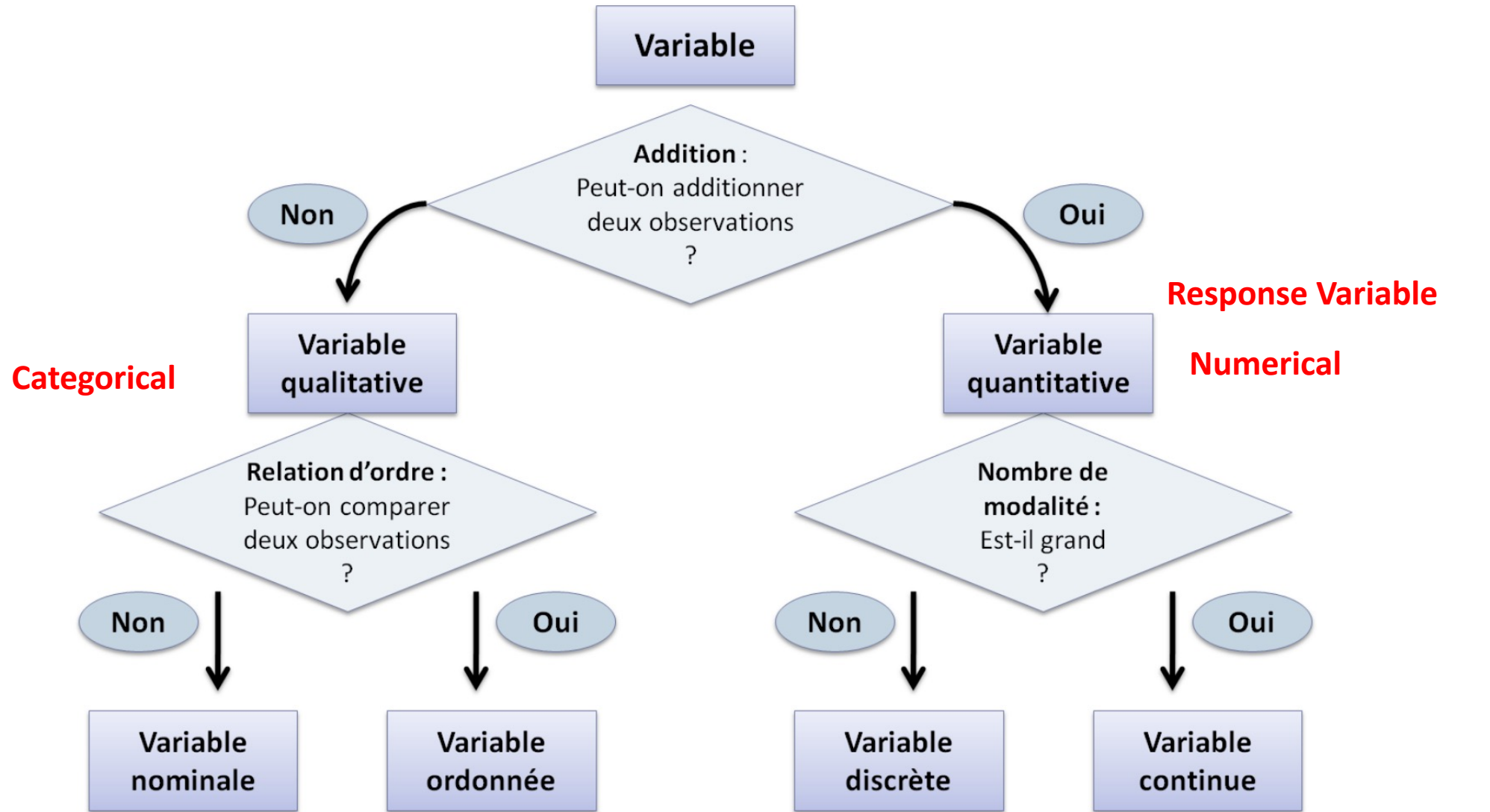
Population

TEST échantillons

	H_0 vraie	H_1 vraie
accepter H_0	OK	erreur type 2 β Faux Négatif
rejeter H_0	erreur type 1 α Faux positif	OK



Reminder on variables... important for statistical tests



- Married, single...
→ No relation order

- Behaviour
- good, excellent...

Child in family (1,2,3..)
finite number of real values

Size, weight : infinite

Bivariate Hypothesis Testing

- Seek to **quantify the association** between a **variable to be explained** (response/Quantitative) and an **explanatory variable** (factor/categorical)
- **Make statistical inferences about the relationship between two variables, One quantitative variable (response) & one qualitative (explicative)!**
 - Can variations in **species richness** (response variable) be explained by the explanatory variable (factor) **Treatment**
 - **Comparison of mean between groups**

- Parametric or non parametric test??
- which test?? significance ? (p-value)
- How many groups??
- Post hoc test required ??



Which test for independent samples?

ONE categorical variable (H/F) & ONE continuous variable (numerical)

Normalité des données?

Shapiro, Q-Q plots

Which test for independent samples?

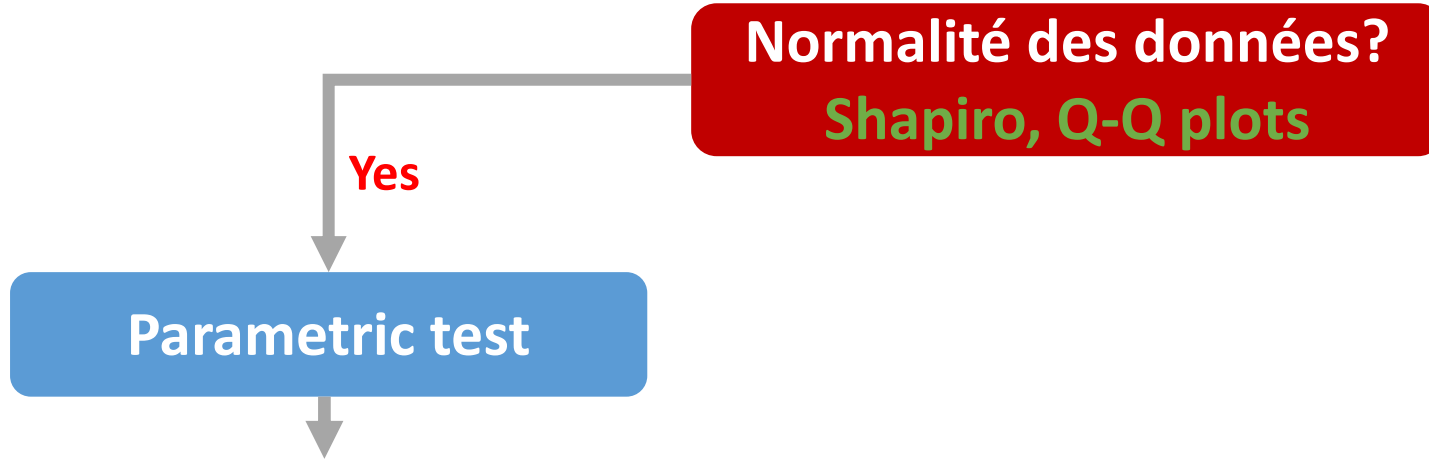
ONE categorical variable (H/F) & ONE continuous variable (numerical)

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Shapiro, Q-Q plots

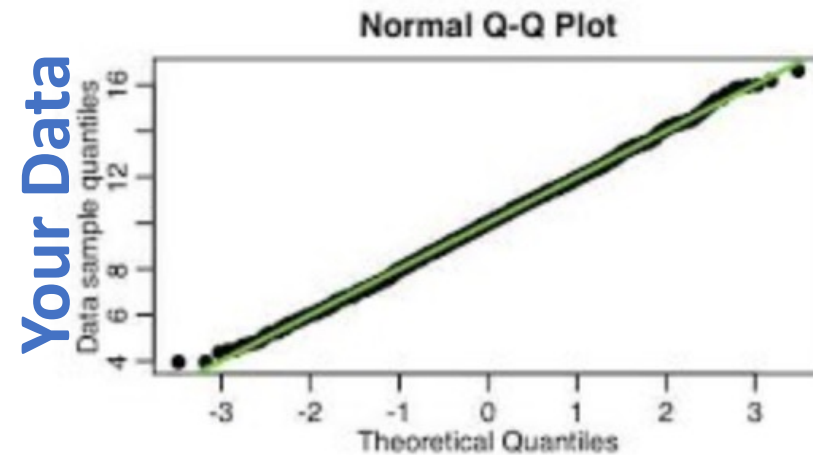
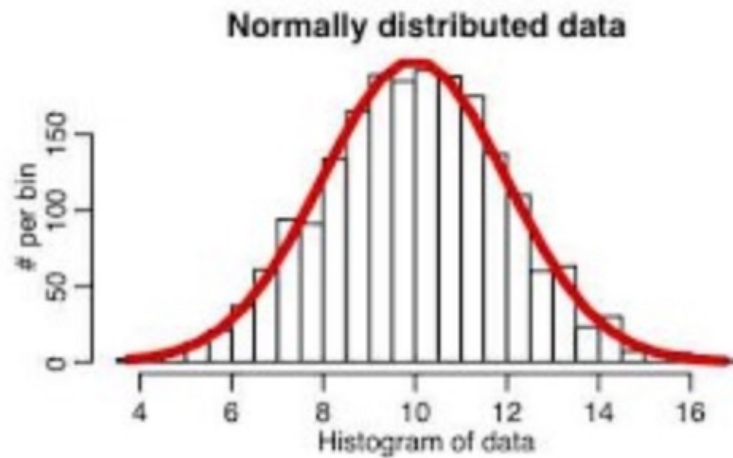
Yes

Parametric test



Q-Q plot normale: Compare your distribution with a normal distribution

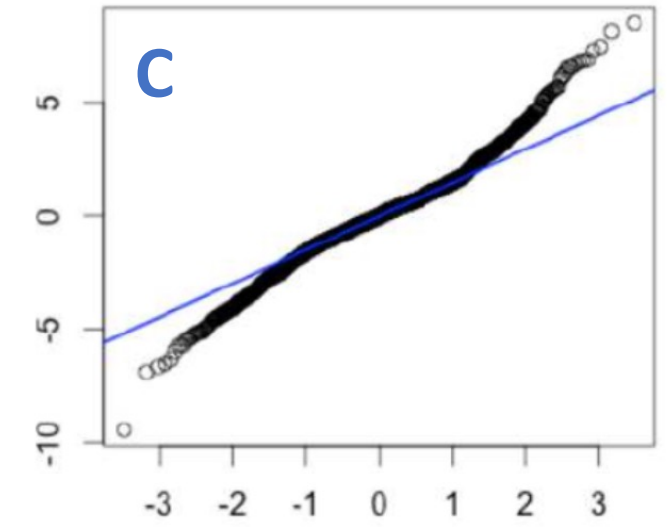
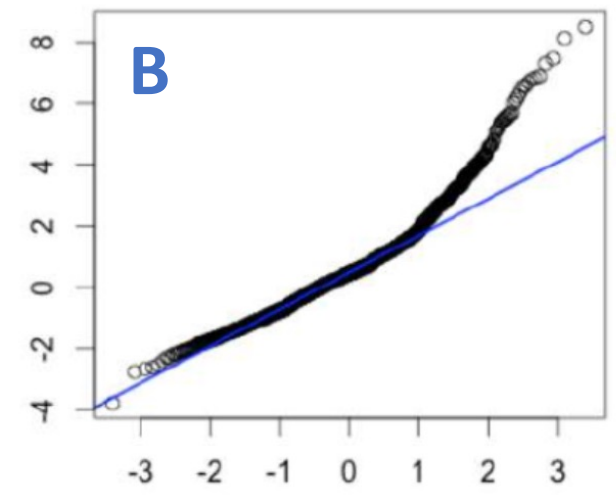
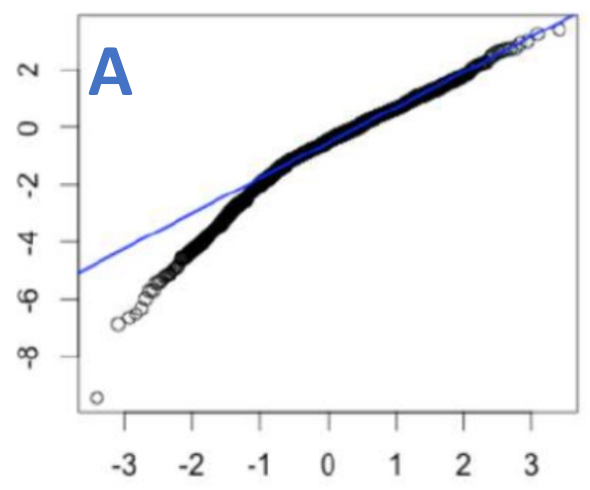
Do my data follow a normal distribution ?



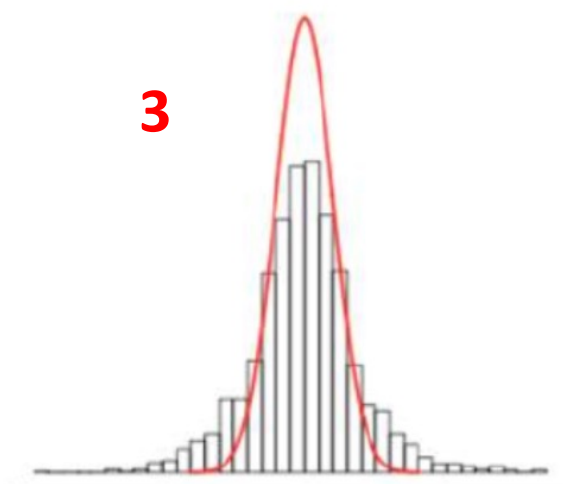
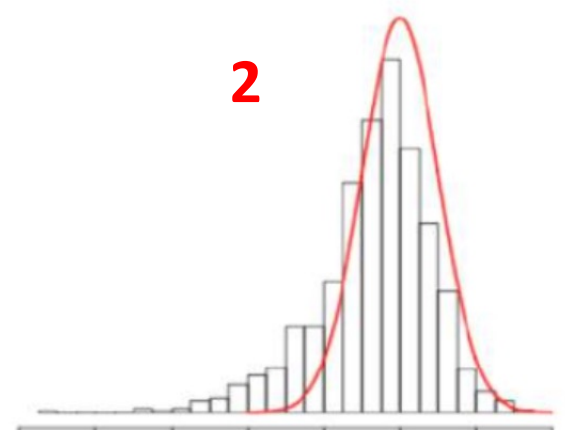
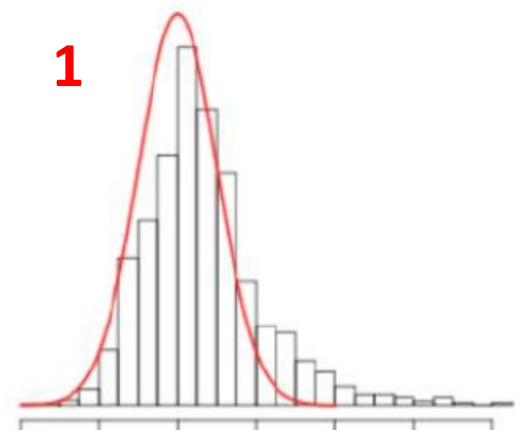
Conclusion?

The line drawn by QQ-Plot indicates the position that the points must have to follow a normal distribution

What are the distributions (bottom) corresponding to these QQ-plots?



??????????



Which test for independent samples?
ONE categorical variable (H/F) & ONE continuous variable (numerical)

Normalité des données?
Shapiro, Q-Q plots

Yes

Parametric test

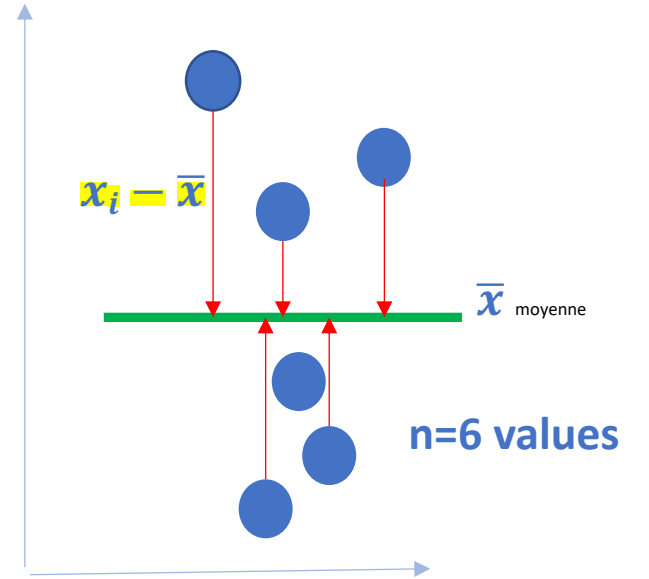
Variance Homogeneity
Bartlett, levene, F-test

Variance= S^2/σ^2

- Variance measures the degree of dispersion of a data set around the mean
- Arithmetic mean of squared deviations from the mean! ☹️

→ square unit

$$S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$



Standard Deviation= S/σ

$S = \sqrt{S^2}$ The advantage of the standard deviation : expressed in the same unit as the data series

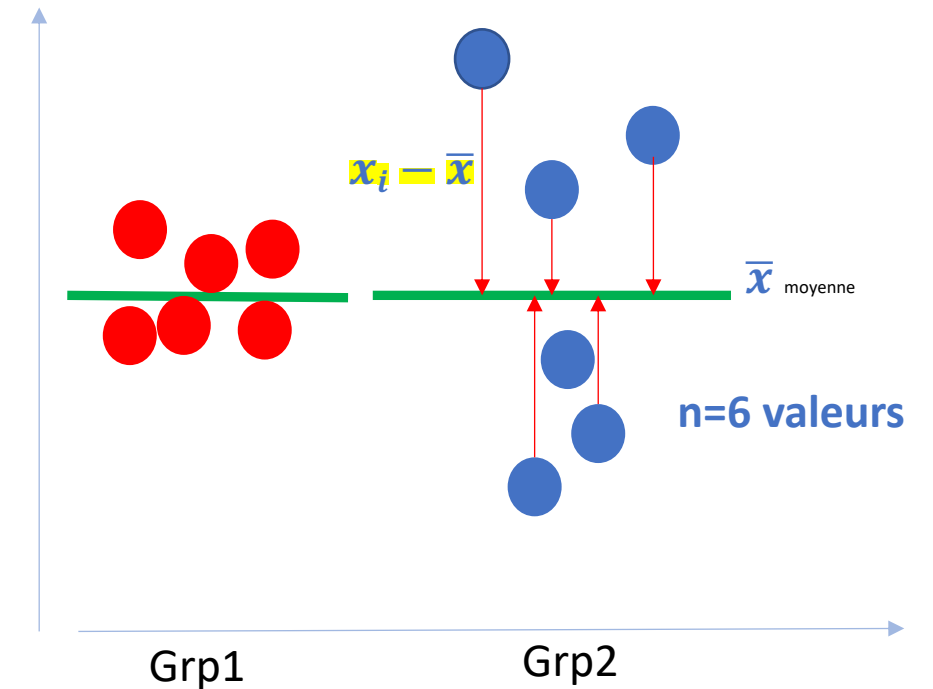
$$S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1} = \frac{\text{Sum of Squares (SS)}}{n-1}$$

SS will be greater in the sample...??

Results of test using variance :

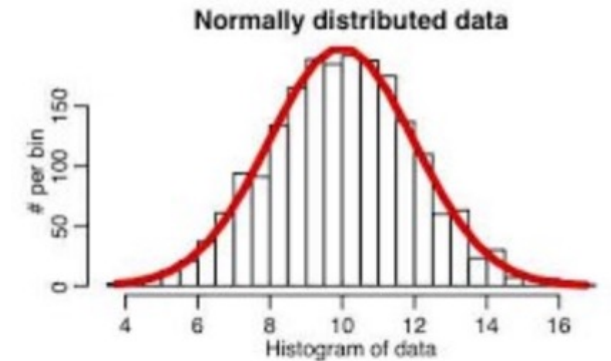
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
groupe	3	13.03	4.343	0.211	0.887
Residuals	14	288.75	20.625		

- *Sum of Squares (= SS, Sum Sq) in your results!*
→ Numerator of variance!!
- *Mean Square (= Mean Sq= VARIANCE formula!!!)*



Requirement for parametric test... check-list!

- Check **normality** of data: Shapiro Test & QQ-plots!!
- Shapiro: H_0 is «data follow normal distribution»



- Check **variance Homogeneity**: F-test (2 groups), Bartlett's & Levene's tests
- H_0 : « No difference »

$$S^2 = 169$$

$$S^2 = 289$$



Parametric Tests

Follow a known distribution (Normal distribution)



Position parameters
Dispersion parameters

Conditions are required (variance homogeneity)

- **T-test (paired or unpaired):** Compare of the means from **2 sample groups** for one variable
- **One way Anova** (variance analysis) : compare the means of **three or more sample groups** for one variable

Which test for independent samples?
ONE categorical variable (H/F) & ONE continuous variable (numerical)

Normalité des données?
Shapiro, Q-Q plots

Yes

Parametric test

Variance Homogeneity
Bartlett, levene, F-test

No

Transformation
(square root, log)

Yes

Yes

How many groups?

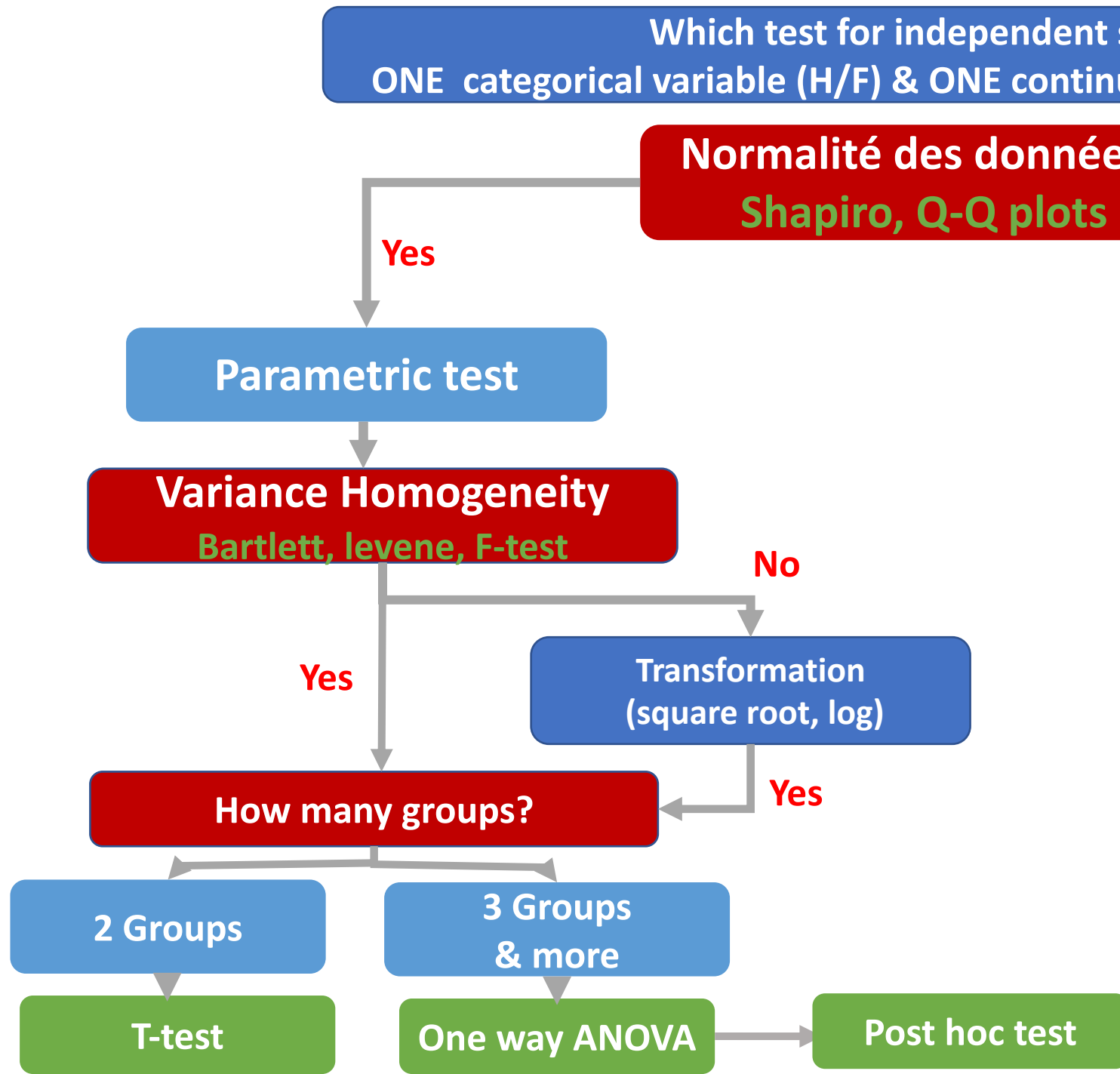
2 Groups

3 Groups
& more

T-test

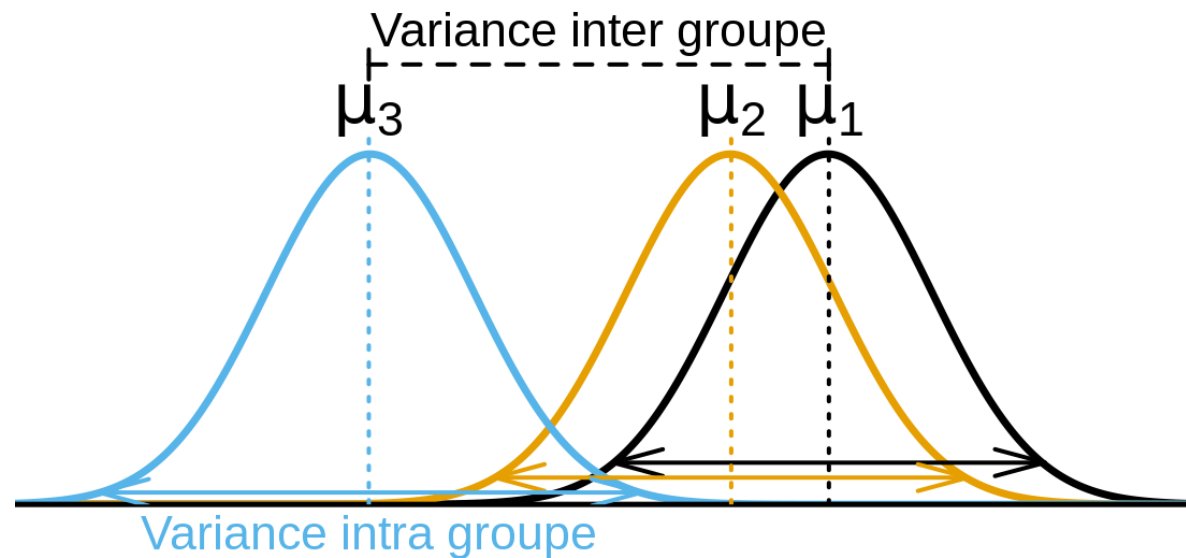
One way ANOVA

Post hoc test



ANOVA: ANalysis Of VAriance (One way Anova= Univariate) (3 groups at least)

- Compare the **variance of the group means** to that **within groups** (i.e. intra-group variance) for a **single explanatory variable** (qualitative)



ANOVA: ANalysis Of VAriance (One way Anova= Univariate)

- Postulate = The **VARIATIONS** observed between the **MEANS** of the different groups (AT LEAST 3) are so small that they are easily explained by chance!!!
- Evaluation : Compare the **variance of the group means** to that **within groups** (i.e. intra-group variance)
- ANOVA → variations through the Variance quantity

Variance inter-groupes + Variance intra-groupes

attribuable au facteur

attribuable à l'expérimentale
(fluctuation de l'échantillonnage, hasard)

Factor effect!

• **Statistic F** =
$$\frac{\text{Inter-group Variance}}{\text{Intra-group Variance}}$$

Chance /fluctuation

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
groupe	3	13.03	4.343	0.211	0.887
Residuals	14	288.75	20.625		

Idea :

if the factor really has an effect, the part of the variations that can be attributed to it = **Inter-group variance** will be significantly higher than the part of the variations that cannot be attributed to it = **Intra-group variance**!

Statistic F Follows a so-called **Fisher-Snedecor** law:

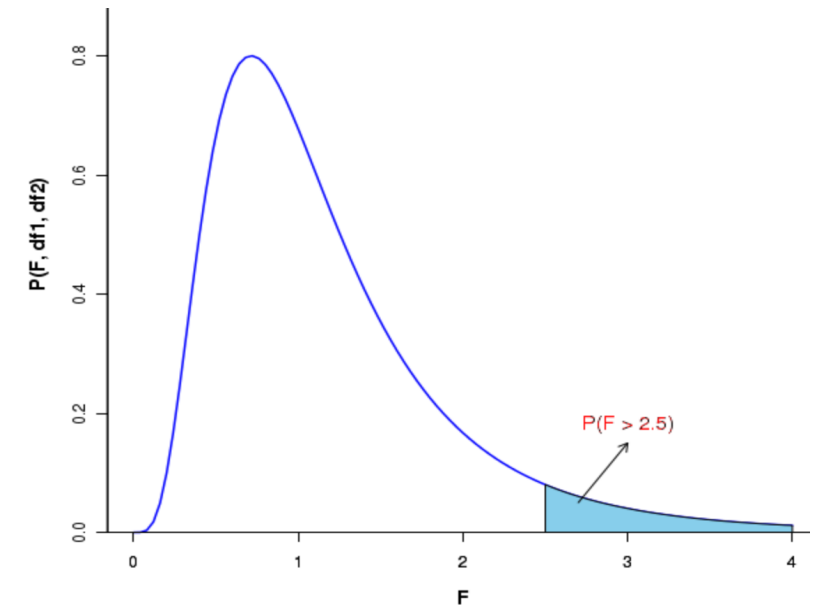
= **Distribution F** used for test of variances, distribution of variances not being normal

- Relation of an observed value of F with the a priori probability of encountering such a value ($>$ or $=$) by chance!
- → probability given by the law = p-value!

	Denominator S^2	Numerator S^2	S^2	
	Df	Sum Sq	Mean Sq	F value Pr(>F)
groupe	3	13.03	4.343	0.211 0.887
Residuals	14	288.75	20.625	

variances	ddl	F
entre k groupes	v_k	$k-1$
résiduelle	v_r	$N - k$

Degré de liberté



- **Two-ways ANOVA : Influences of TWO qualitative variables on ONE quantitative variable**

Exple: Influence of soil type and degree of humidity (ordinal variable) on plant yield

Non-parametric tests

No assumptions are made for the distribution of data:
Distribution-free tests, they are alternative to parametric tests

- Wilcoxon Rank test : samples are paired/unpaired, 2 sample groups
- Mann-Withney test: Independent samples, 2 sample groups
- Kruskal wallis test : Independant samples, Three or more groups

→ Based on the average ranks: we classify the values, we replace by a position (1,2 etc),
Compares the average of the ranks between the groups

Which test for independent samples?
ONE categorical variable (H/F) & ONE continuous variable (numerical)

Normality of data?
Shapiro, Q-Q plots

NO

Non parametric test

How many groups?

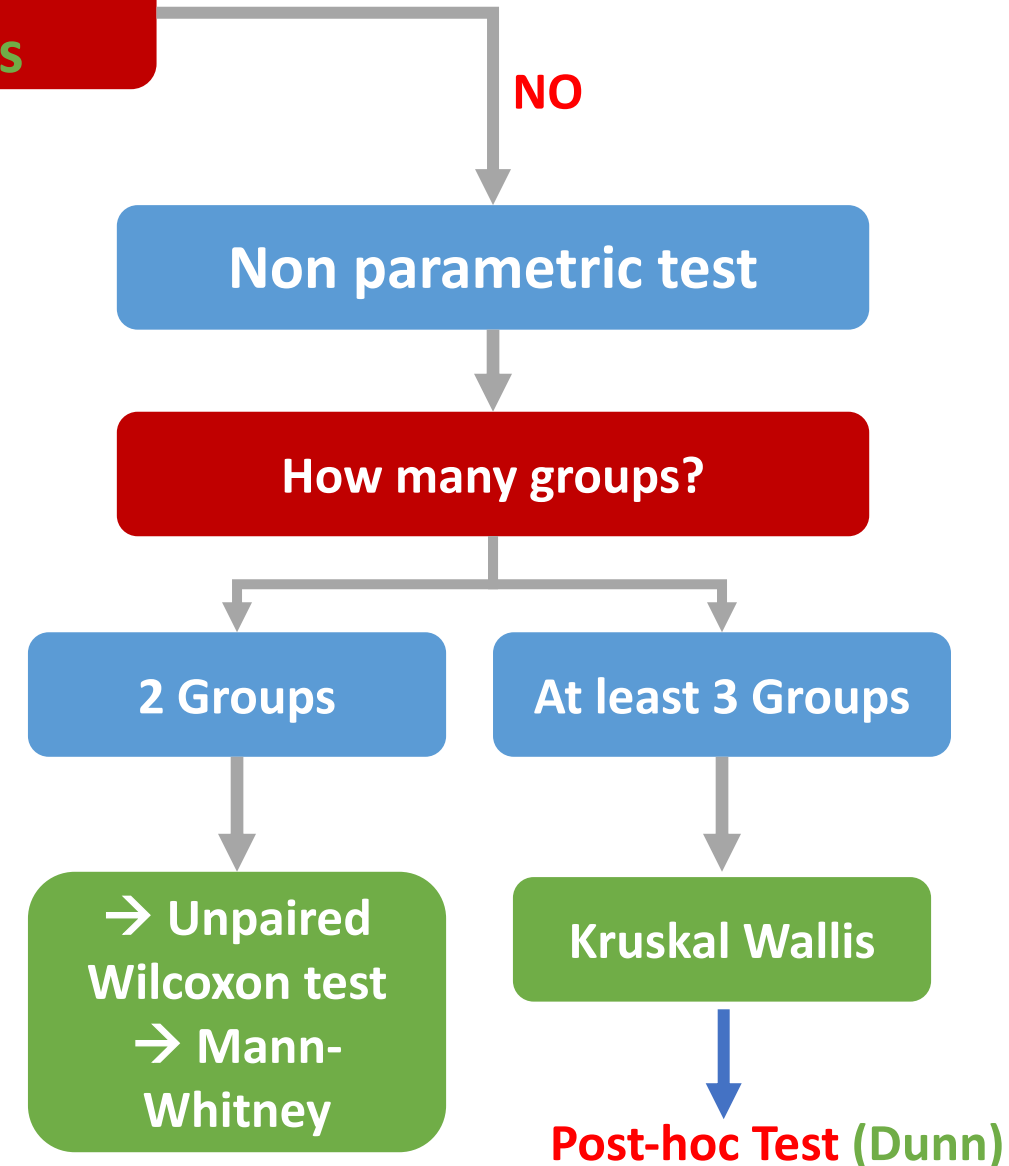
2 Groups

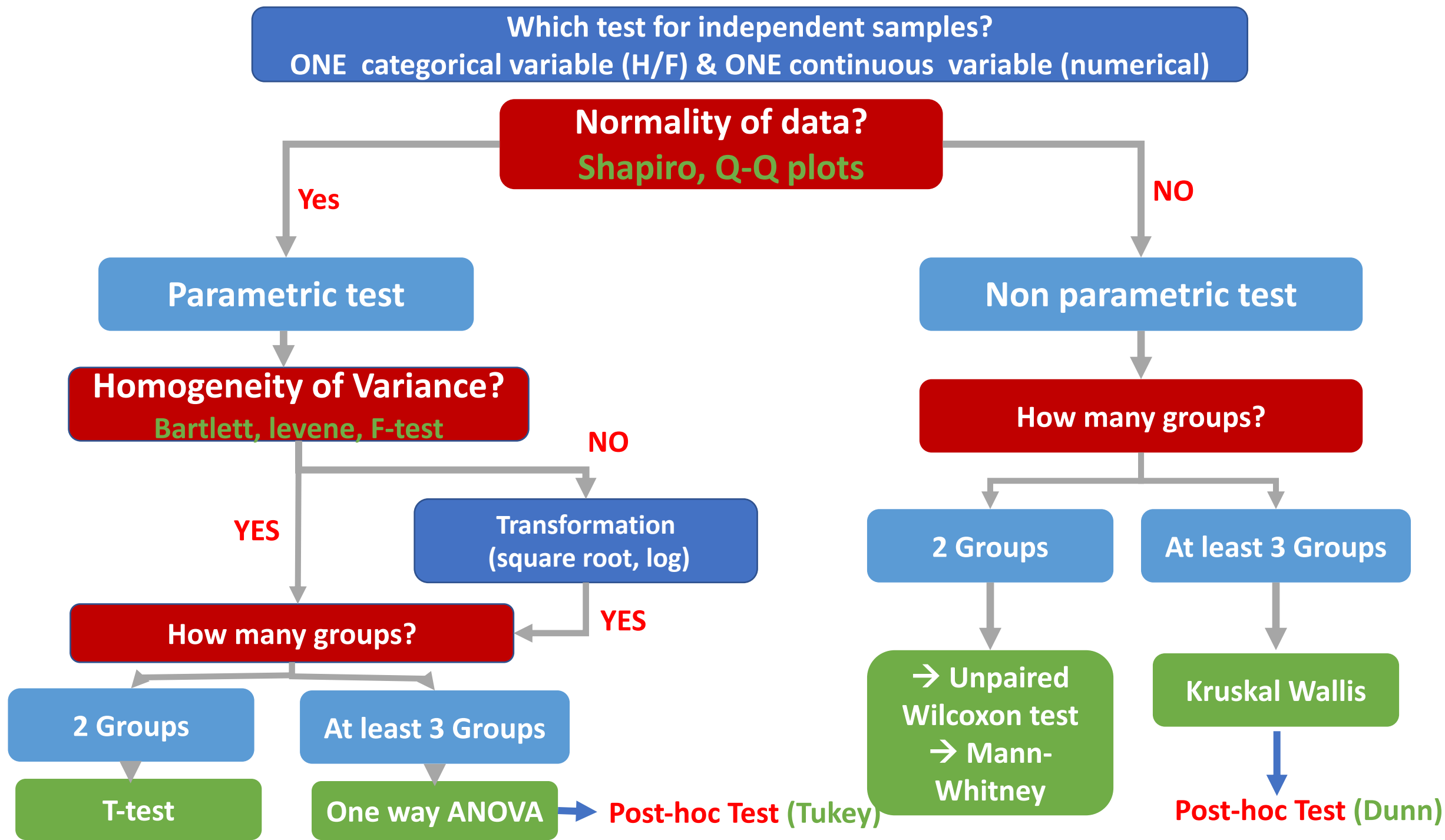
→ Unpaired
Wilcoxon test
→ Mann-
Whitney

At least 3 Groups

Kruskal Wallis

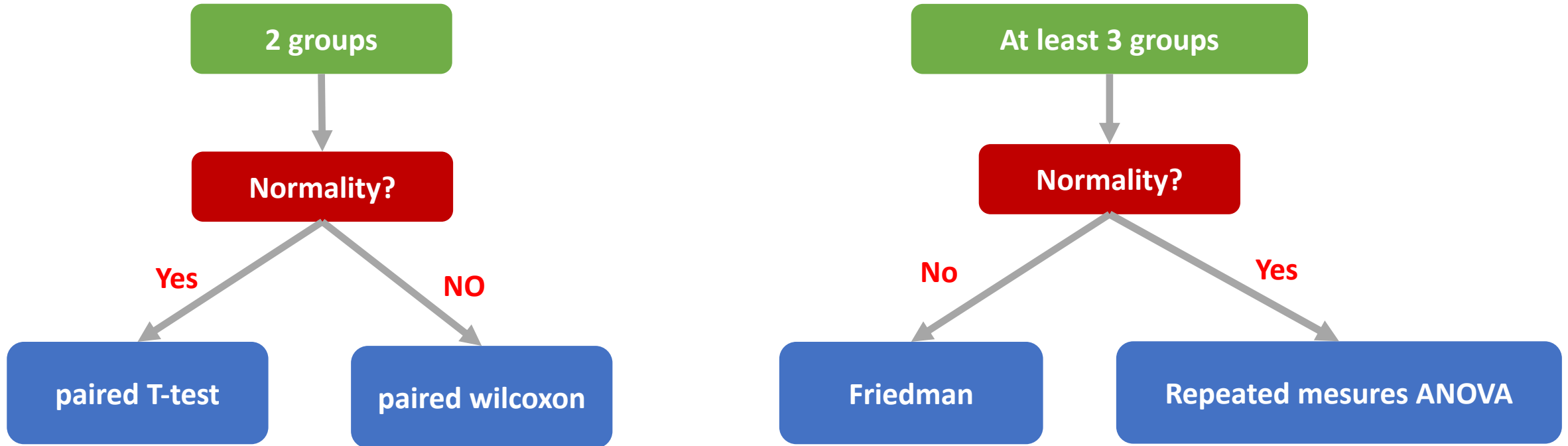
Post-hoc Test (Dunn)





Repeated measurements – paired samples

Exple= time series, Before-After
Treatment...



Post-hoc Test

Statistical tests with **at least 3 groups!**

After ANOVA, Kruskal-wallis

→ The result of an ANOVA test is **an Overall p-value**

Exple: You are comparing the effect of 3 soil types (A,B,C) on plant growth

ANOVA returns a p-value of 0.03

It does not tell you which pair of groups are significantly different!!!!

→ Post-hoc Test! Multiple comparisons (eg: Grp A vs. Grp. B; GrpB vs. Grp C; Grp C vs. Grp A!)

- Parametric Post-hoc test (ANOVA) → **Tukey Test**
- Non-parametric Post-hoc test (Kruskal wallis) → **Dunn Test**

Multiple Testing Issue: increasing the risk...

Test is based on **probabilities**, so there is always **a risk** of drawing the **wrong conclusion!**

→ **No hypothesis test is 100% reliable**



Performing hypothesis testing:

- You have two hypotheses :
 - H_0 : Null hypothesis = the reference hypothesis : No difference
 - H_1 : Alternative hypothesis: There is a difference

- You encounter: **Type I error : α = Risk alpha**

$\alpha = 0.05$ Is the **probability** (significance threshold) to incorrectly **reject H_0 !**

In other words, an acceptable chance of a false positive!!

Differential abundance : Multiple testing!!

ONE TEST :

$$P_{\text{False Positive}} = P_{\text{error}} = \underline{\alpha} = 0.05$$

Complementary Prob

$$P_{\text{no_error}} = 1 - \underline{\alpha} = 0.95$$

TWO TEST without making error : $P_{\text{no_error in two tests}} = (1 - \underline{\alpha}) * (1 - \underline{\alpha}) = (1 - \underline{\alpha})^2$

Complementary Prob

$$P_{\text{at_least_ONE_error in two tests}} = 1 - (1 - \underline{\alpha})^2$$

Generalization to n TESTS

$$P_{\text{at_least_ONE_error in n tests}} = 1 - (1 - \underline{\alpha})^n$$

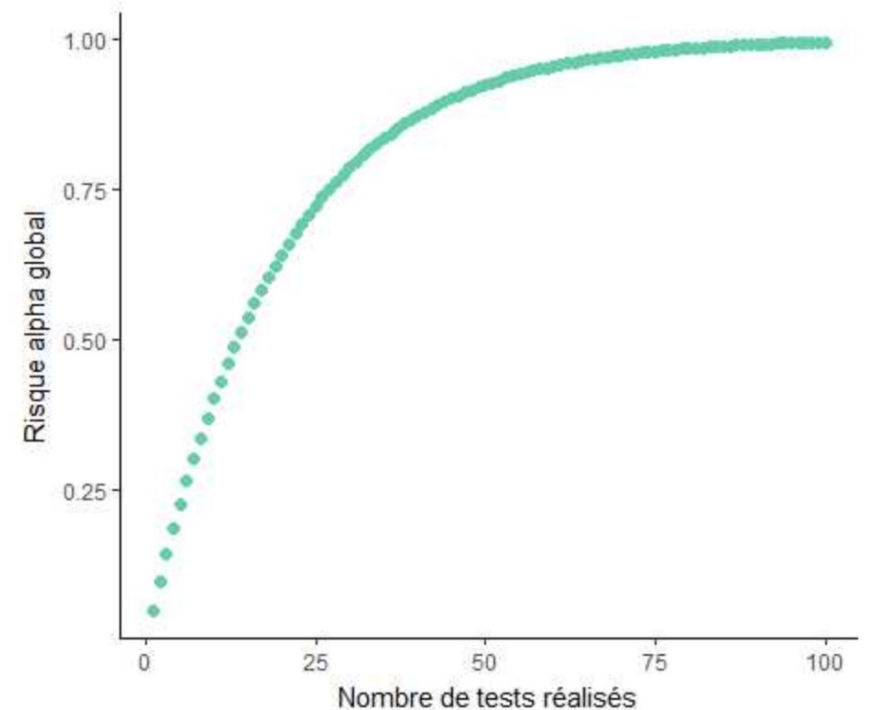
It's called the global α risk

What does it means...

- You test **ONE** ASVs ($n=1$) for differential abundance: $1-(1-\underline{\alpha})^n = 1-(1-0.05)^1 = \mathbf{0.05}$
- You test **3** ASVs ($n=3$): $1-(1-0.05)^3 = \mathbf{0.14}$
- You test **100** ASVs ($n=100$): $1-(1-0.05)^{100} = \mathbf{0.9941}$

The global risk $\underline{\alpha}$ reach $0.9941=99.41\%!!!!$

→ 99% to wrongly reject the H_0 at least
One times



Need to ajusted this phenomen by using p-value **adjusted!**

FDR : False Discovery Rate : Benjamini-Hochker

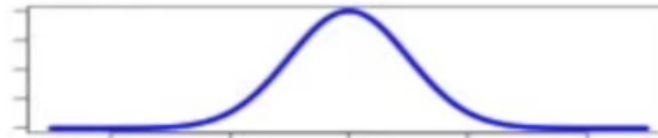
The idea : Discard bad data that looks good!!!

Benjamini-hocherk **adjusts p-values**
to limit the number of **false positives**
that are reported as **significant** (pvalue < 0.05)

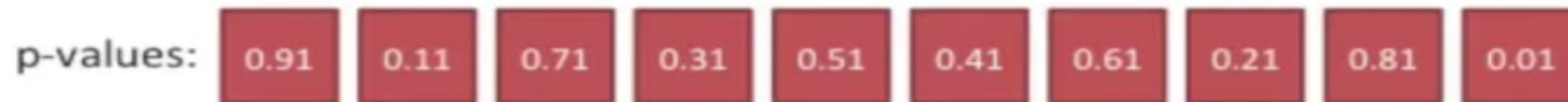
Adjusts p-values
means that it makes them **larger!**

Using FDR cutoff < 0.05
means less than 5% of the significant results will be false positives

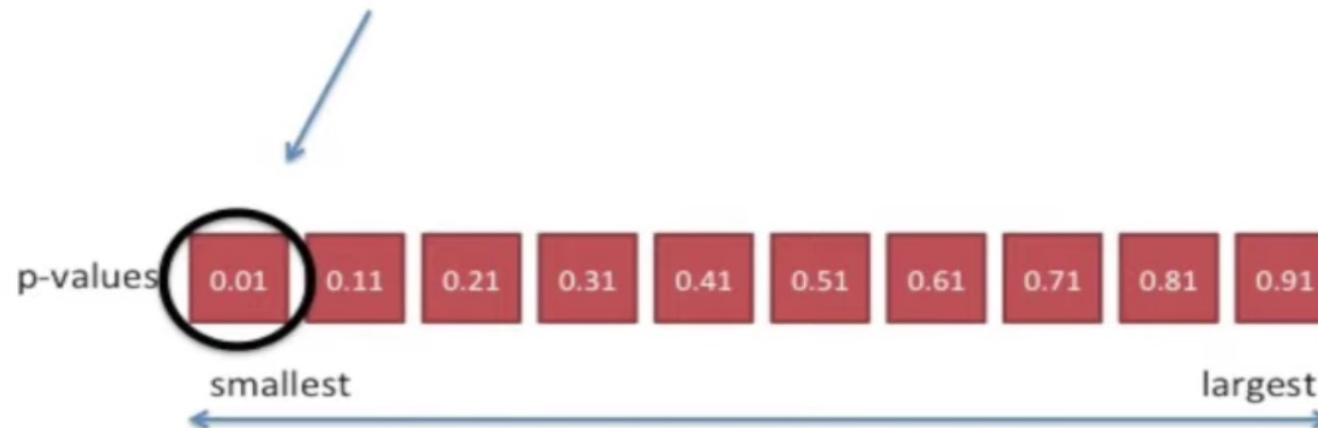
Mathematical approach FDR-Benjamini-Hochker



10 pairs of samples taken from the same distribution. (i.e. 10 genes that were not effected by the drug).

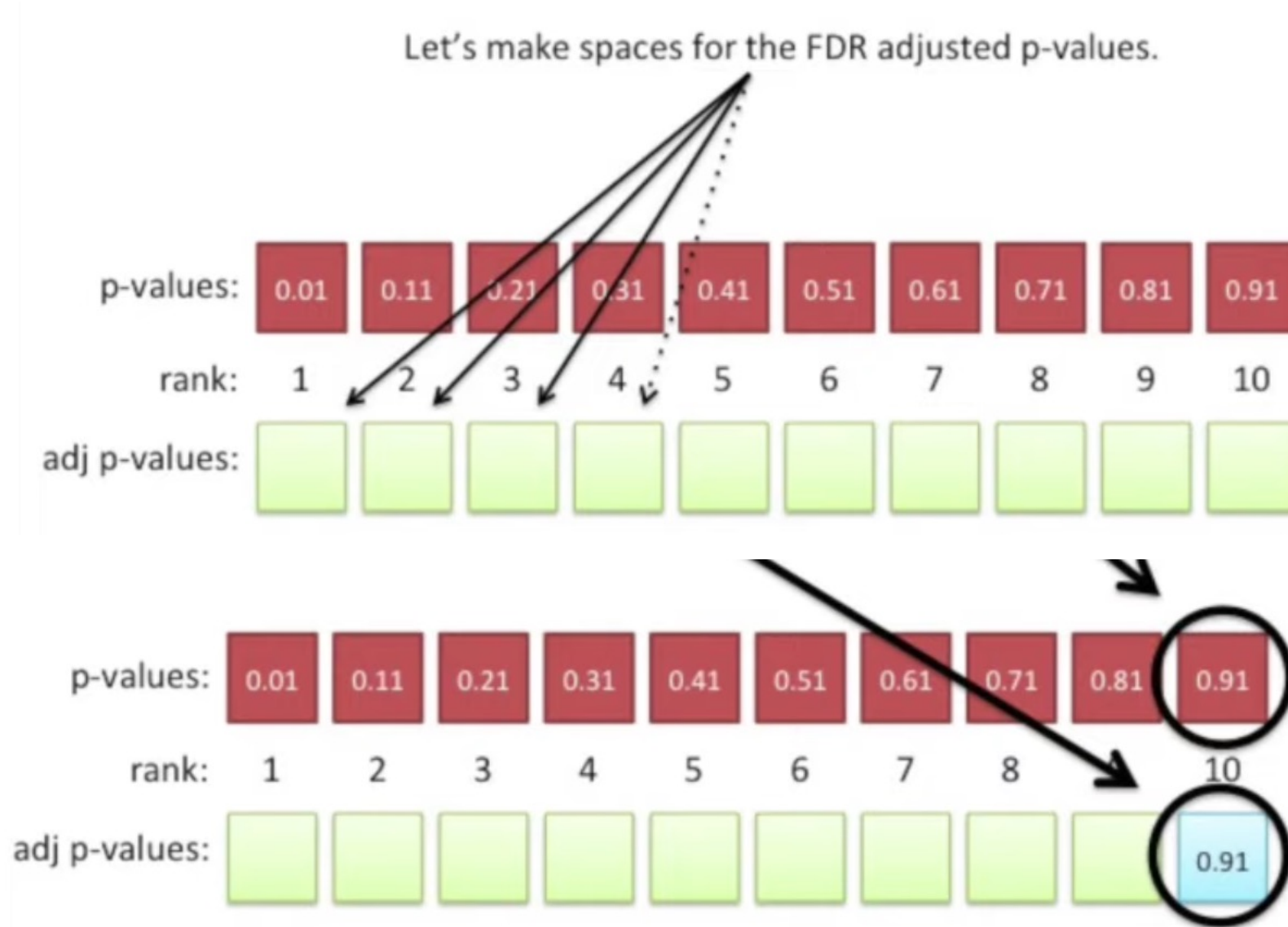


Notice that one of the p-values is a false positive (that is to say, less than 0.05)



1- Ranking pvalue

Prepare space for adjusted p-value



2- Largest adjusted pvalue and larger pvalue are same

Next adjusted pvalue ...

[illegible]

The smallest of the two options

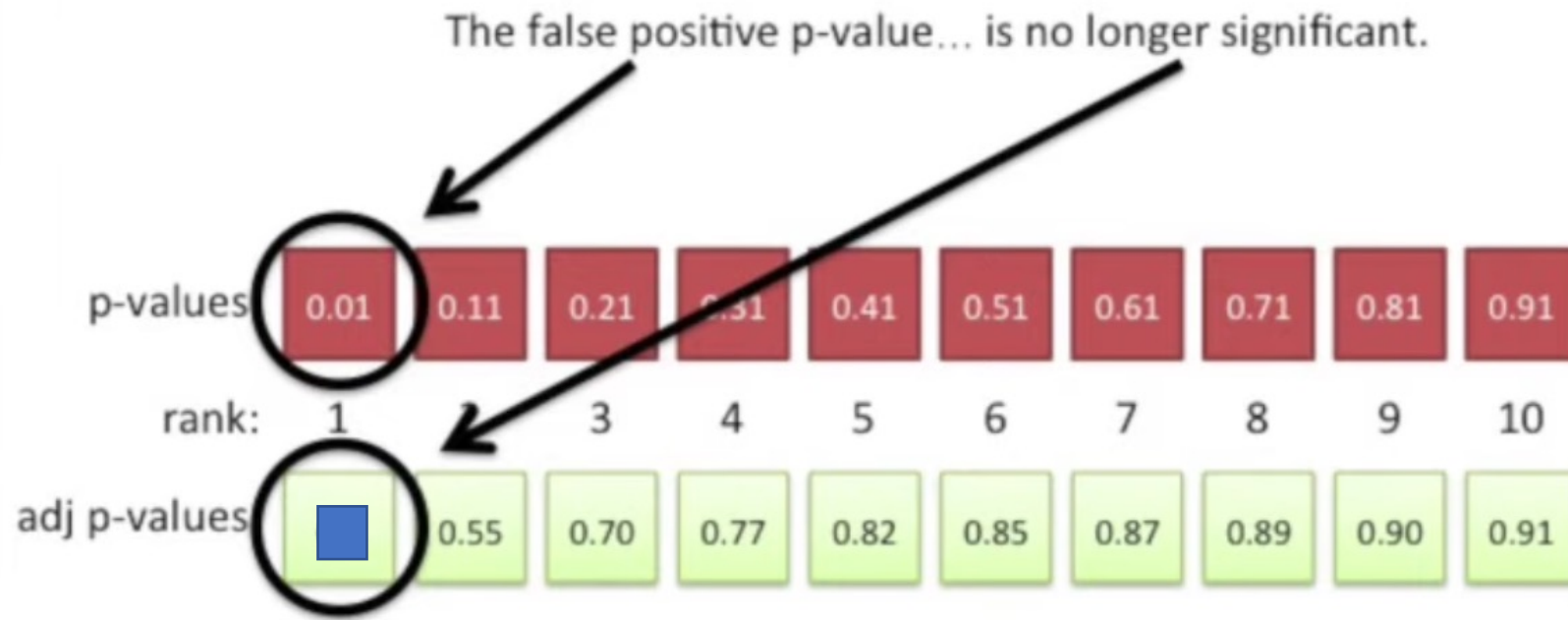
b: the current p-value * $\left(\frac{\text{total \# of p-values}}{\text{p-value rank}} \right)$

b: 0.81 * $\left(\frac{10}{9} \right) = 0.90$

a: The previous adjusted p-value = 0.91

[illegible]

Finally...



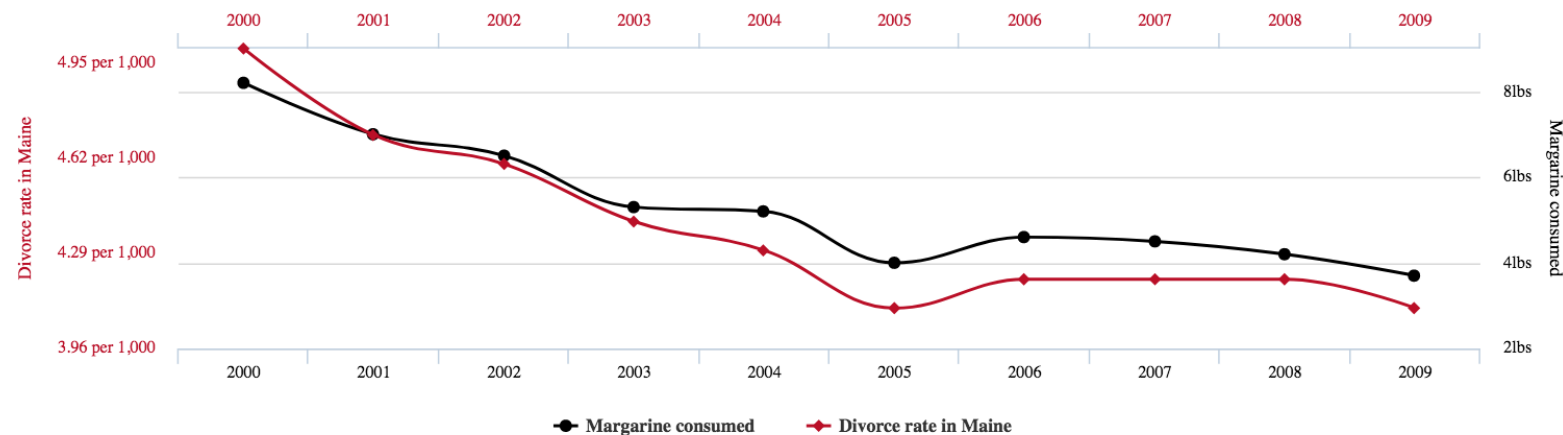
Correlation & linear Regression as Bivariate Analyses

Objective : Analyze the **link** that may exist between **two variables** (here: quantitatives)
(Two qualitative variables -> Khi2 test)

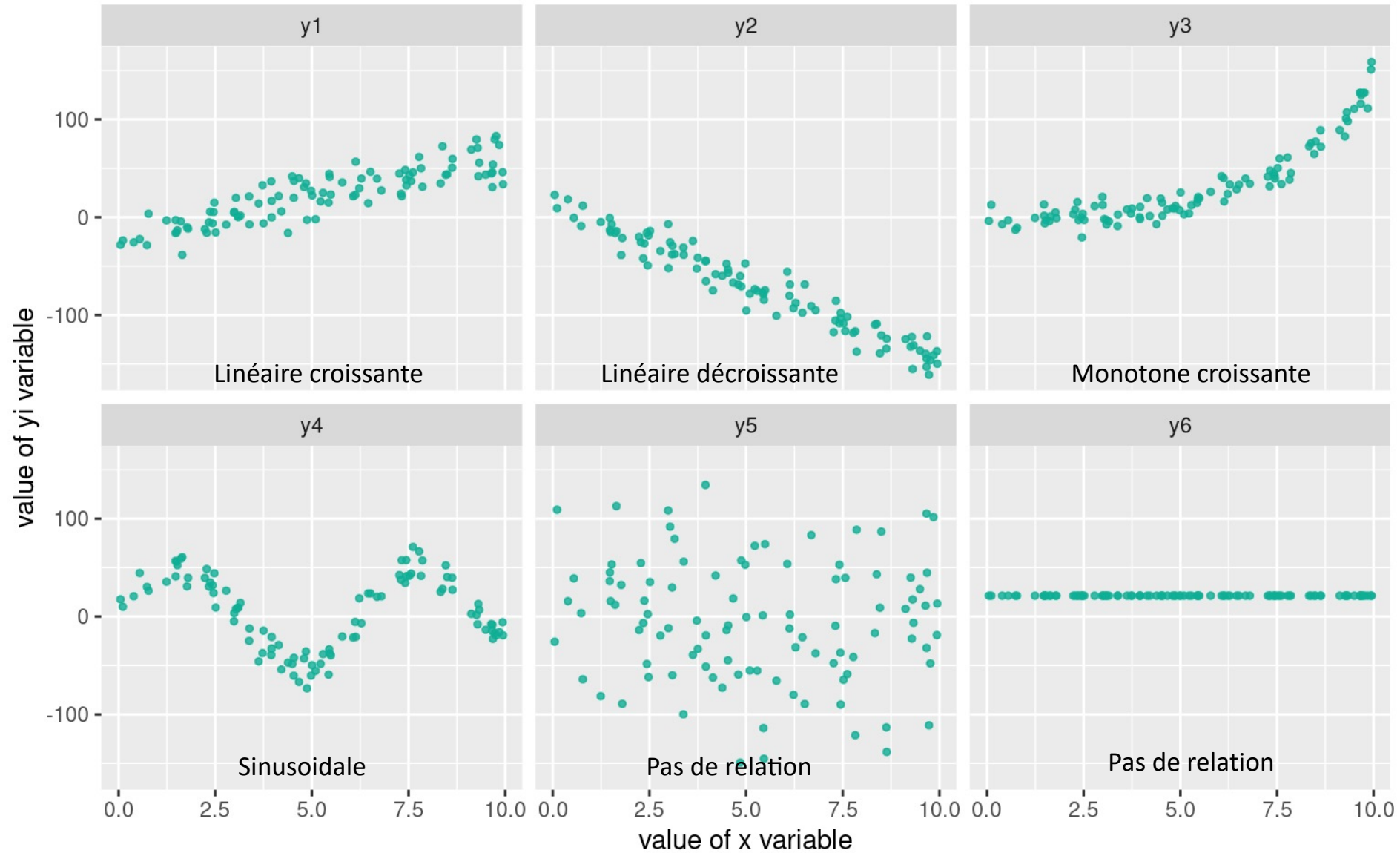
Link/relationship/dependence between the variables

→ The values of two variables **do not evolve independently** but on the contrary, present a certain form, a certain regularity

→ Intensity of the association does not indicate causality ...



What are the relationship between the variables in each graph?

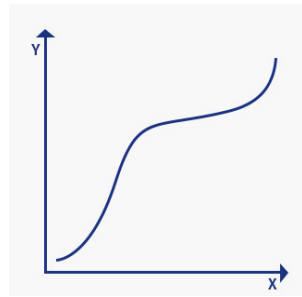


About Correlation Coefficient r

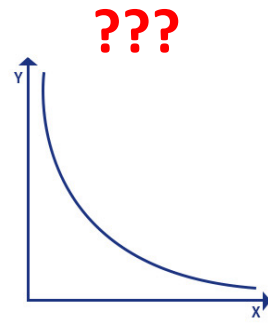
Intensity & Direction of the association between two variables

Methods available:

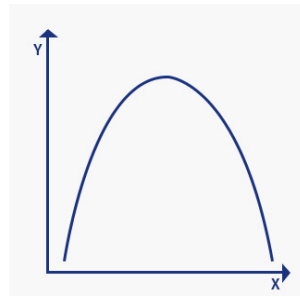
- **Strict Linear Relationship** : **Pearson** (**r**, parametric)
- **Monotonous relationship**, non-parametric : **Spearman** (**Rho**, rank-based)
Kendall (**Tau**), Alternative to Spearman (small sampling)



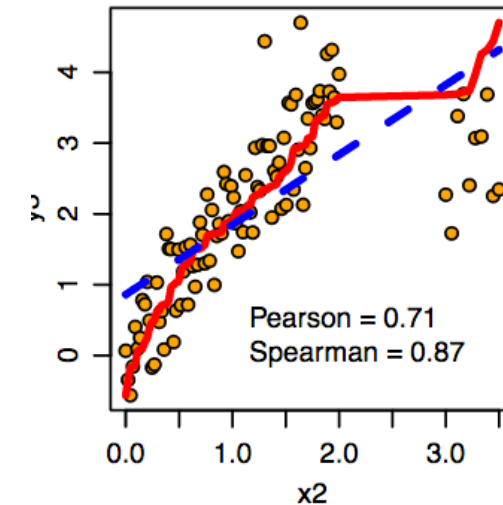
Monotone
croissante



Monotone
décroissante



point d'inflection :
Unimodale

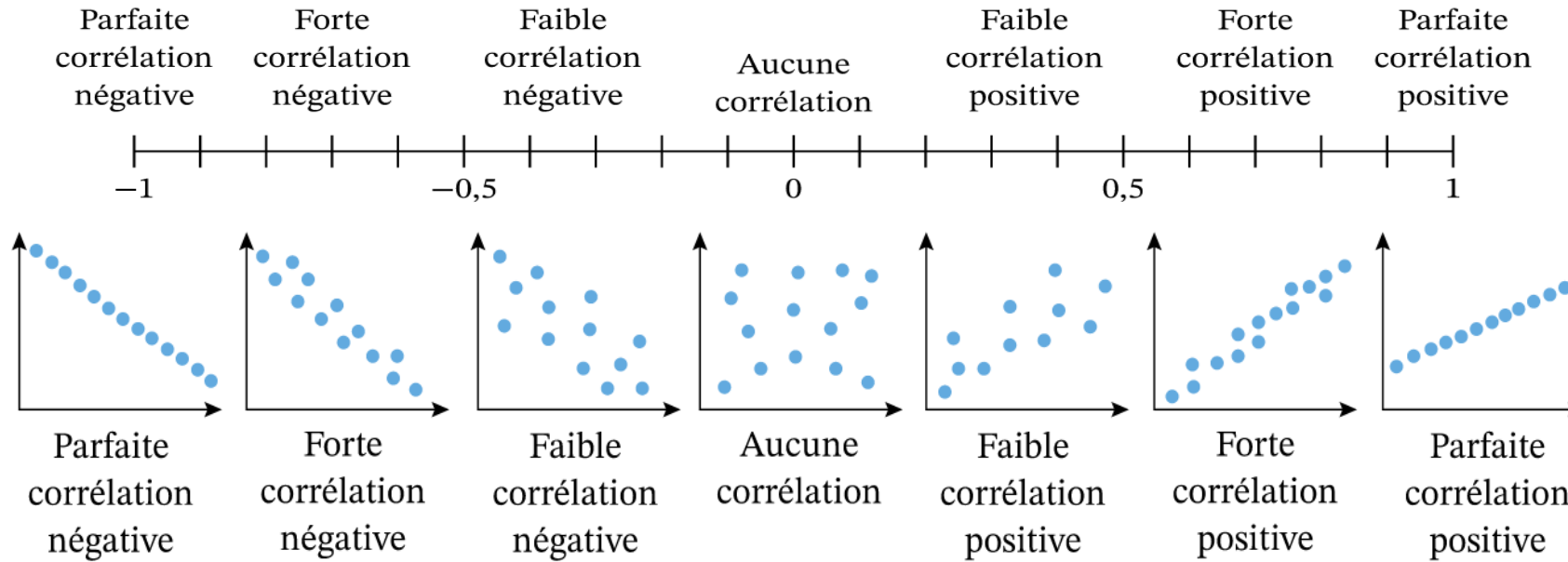


Coefficient r range between -1 et 1

- **Positive correlation** : The values of both variables tend to increase together
- **Negative correlation** : The values of one variable tend to increase and the values of the other variable decrease
- **Zero** : no **LINEAR** association (**Pearson**)

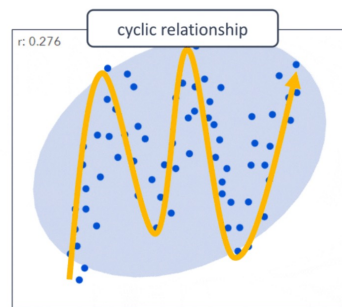
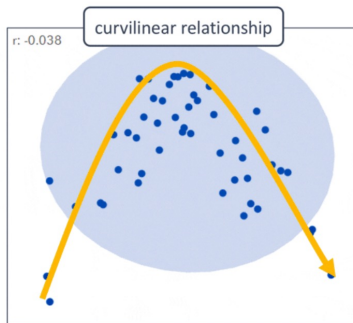
Distribution law of r under the H0 hypothesis (No statistical link between X and Y) → Access to p-values to check significance

For information!!!

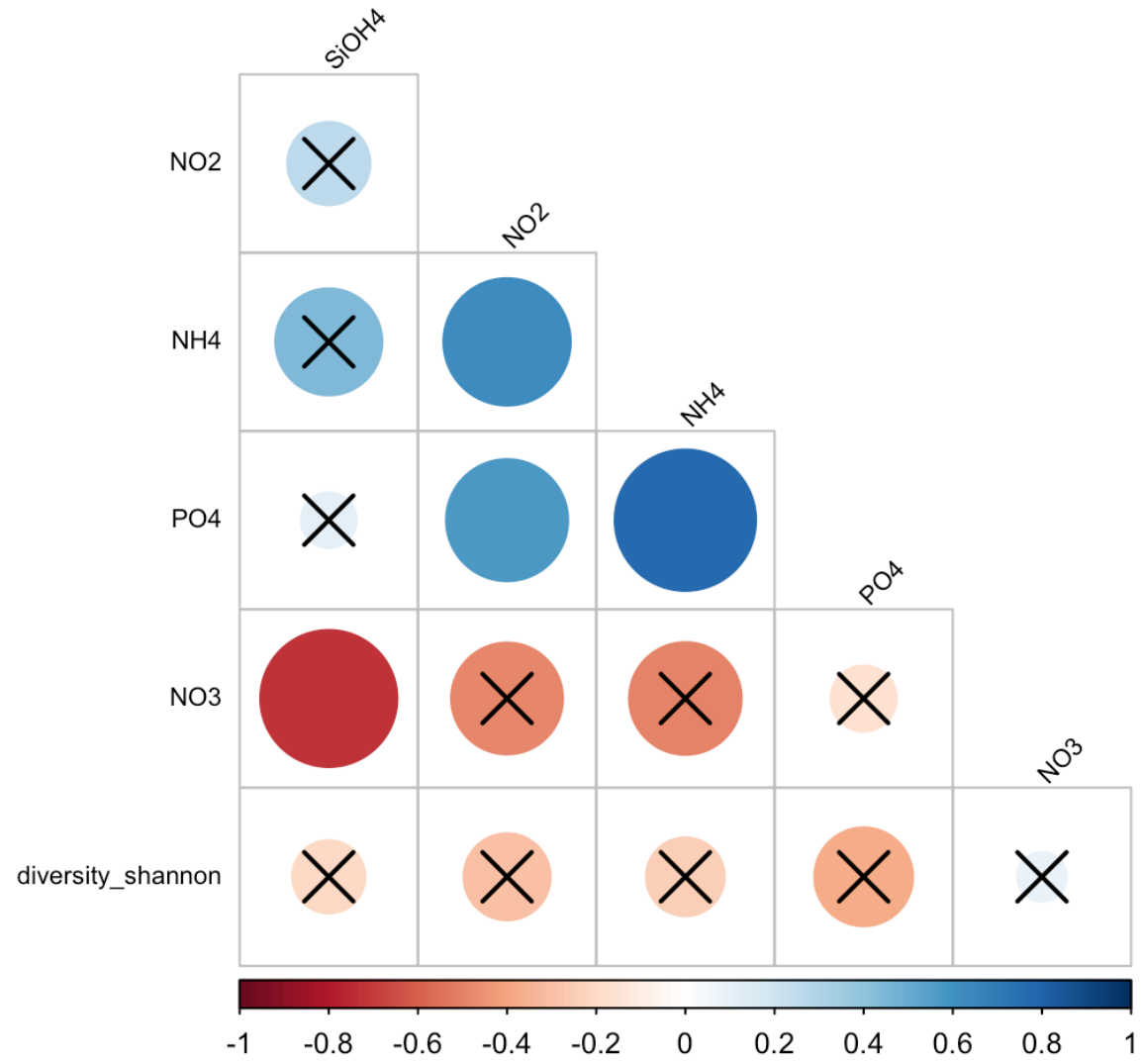


Because inspecting your results graphically is never useless...

- r close to Zero: no association??



Example of bivariate correlation plot
(with significant test)



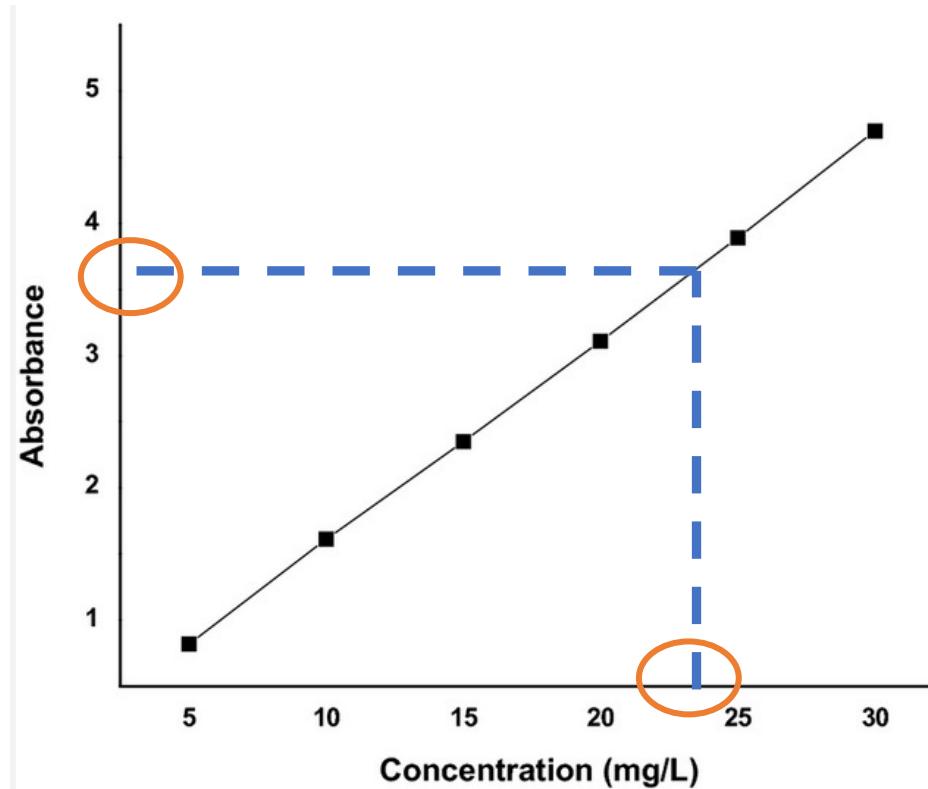
About simple Linear Regression

- Only for quantitative variables
- Plot the scatter plot Is there a relationship?
- Is it linear?
- What orientation (positive, negative)?
- If the association is linear → Make a regression

Requierement

- Normal distribution
- Variance homogeneity

Your favorite linear regression... calibration curve!!!

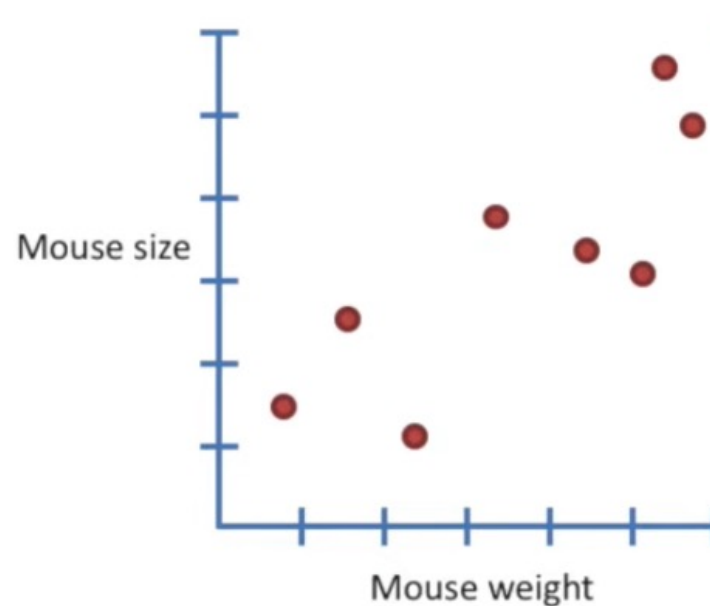


Explain and predict!

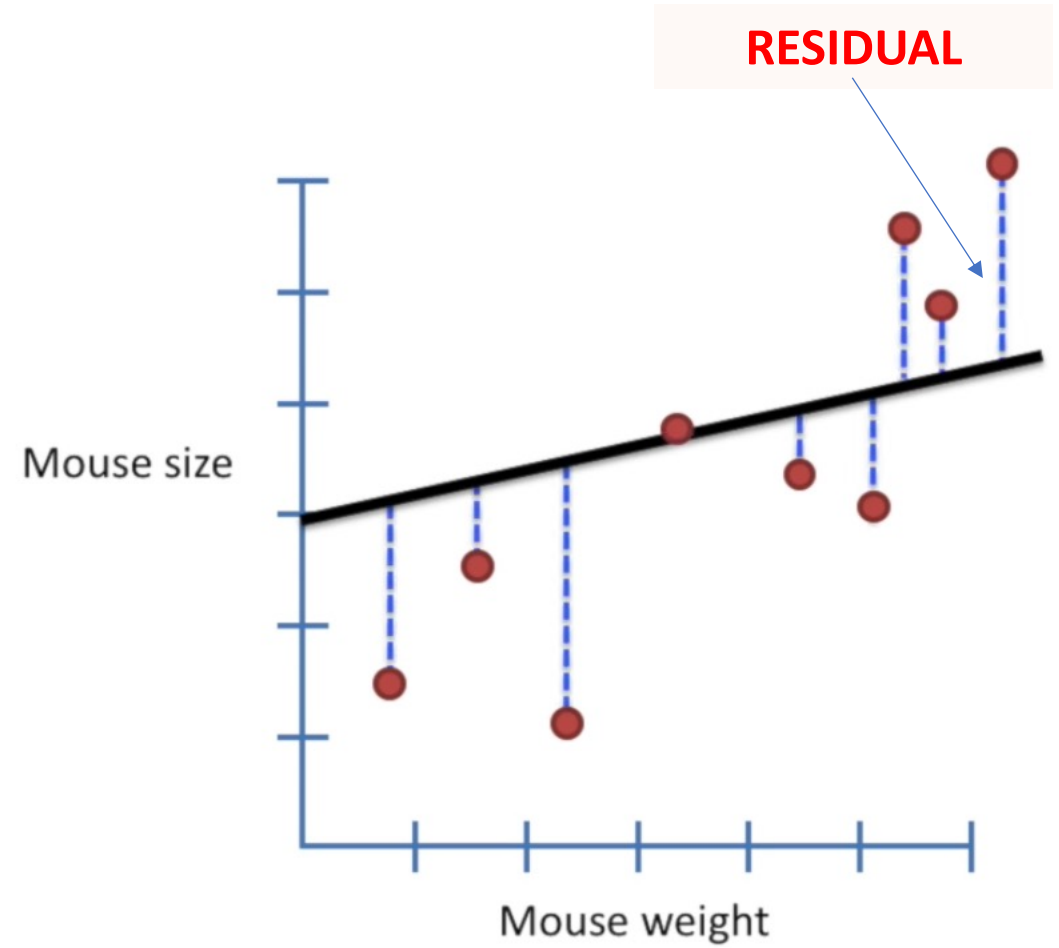
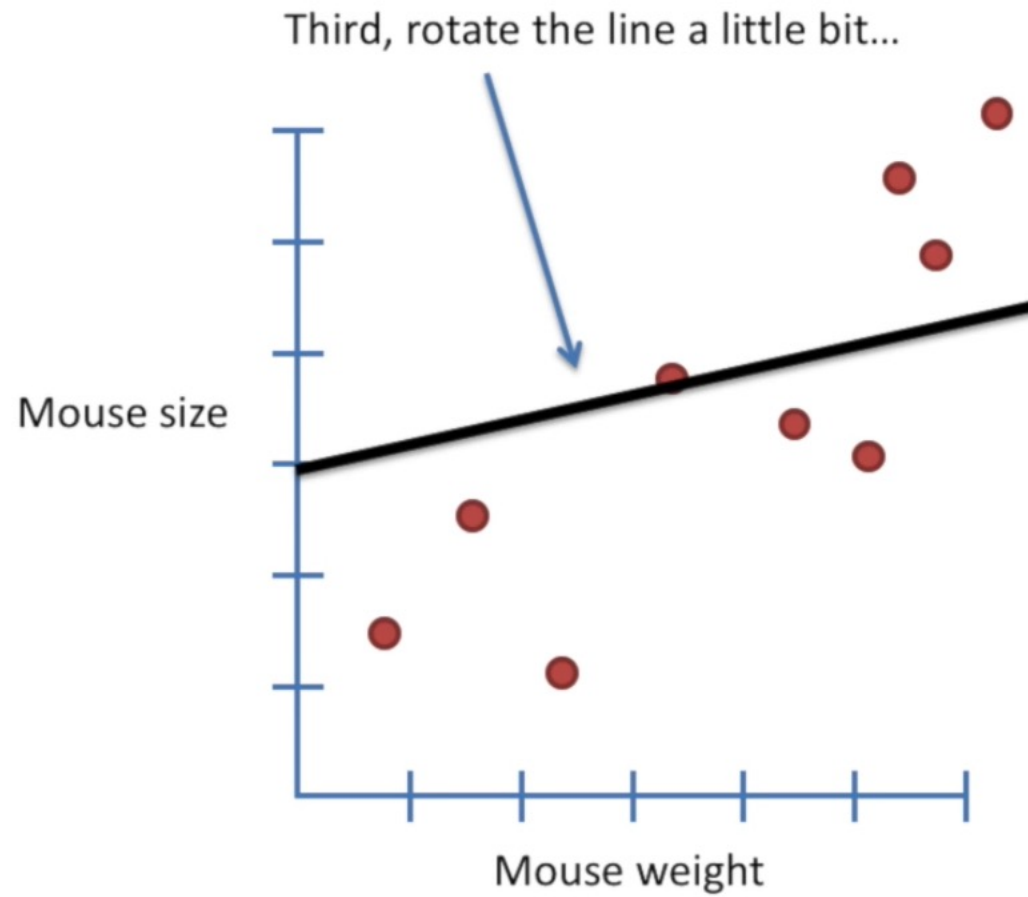
Models a linear type relationship ($Y=aX+b$)

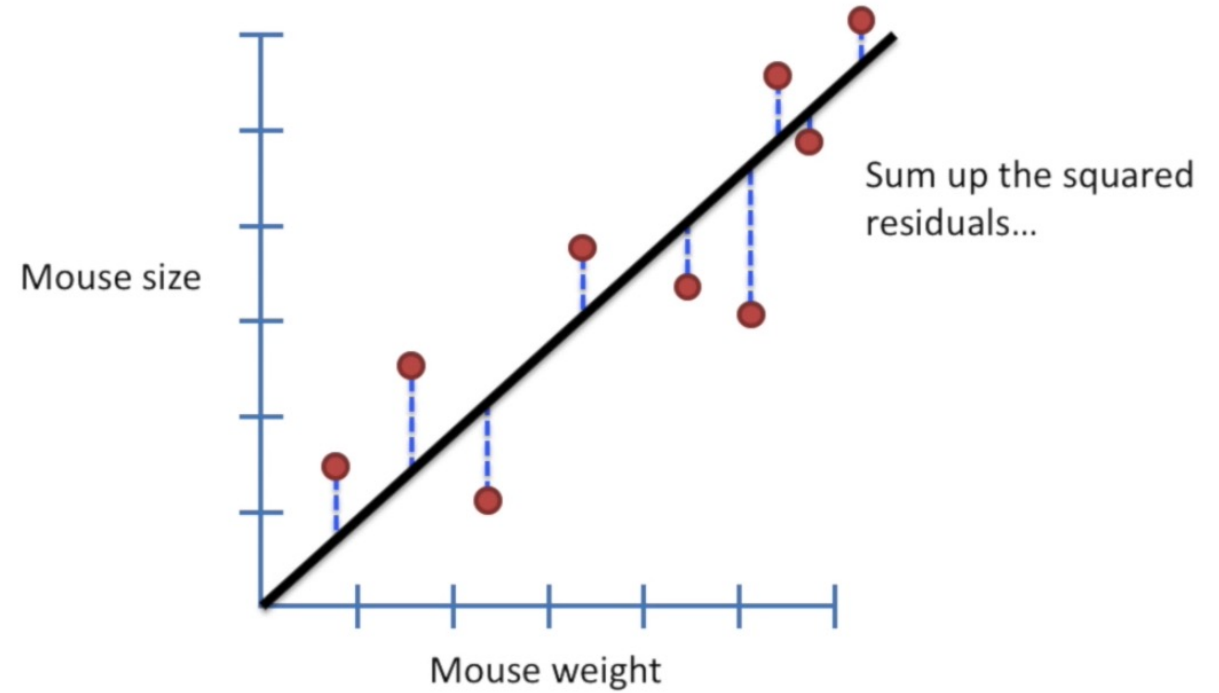
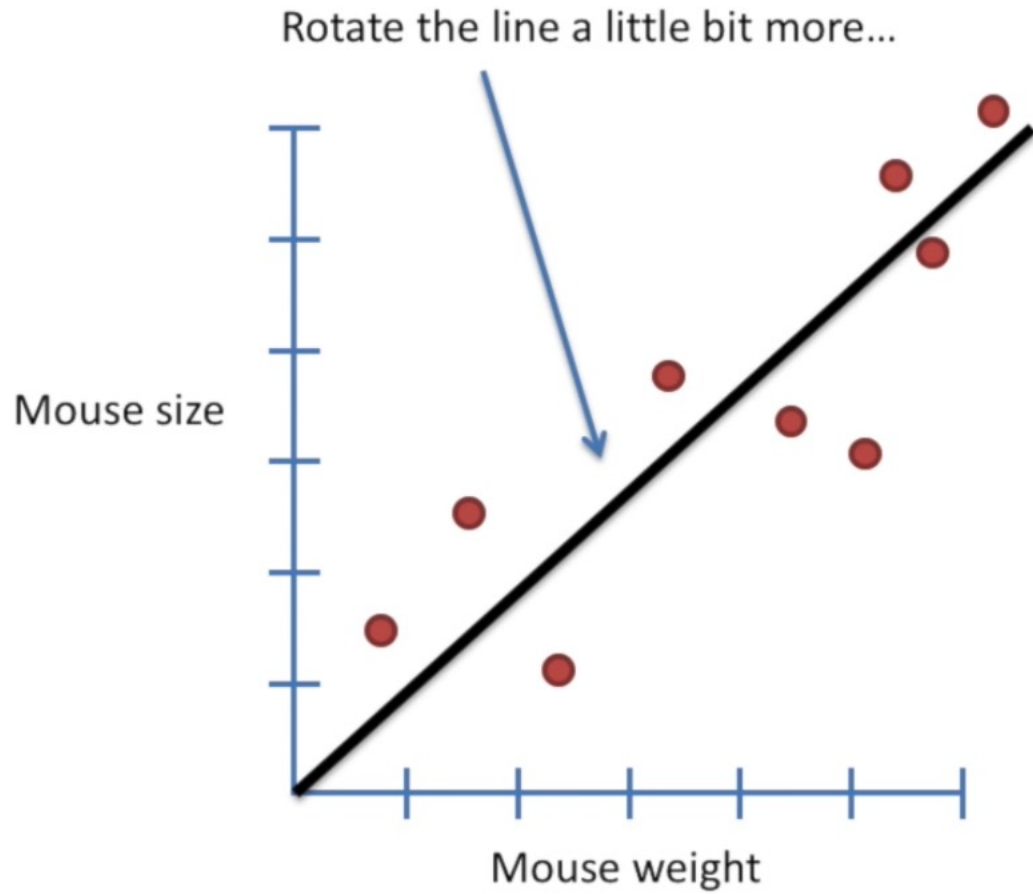
Model seeking to establish a **linear relationship** between a variable, called **explained/dependent (Y)**, and another called **explanatory/independent (X)**

Can mouse **Weight predict **Size** correctly? (R^2)**
Relationship is due to chance? (p-value)



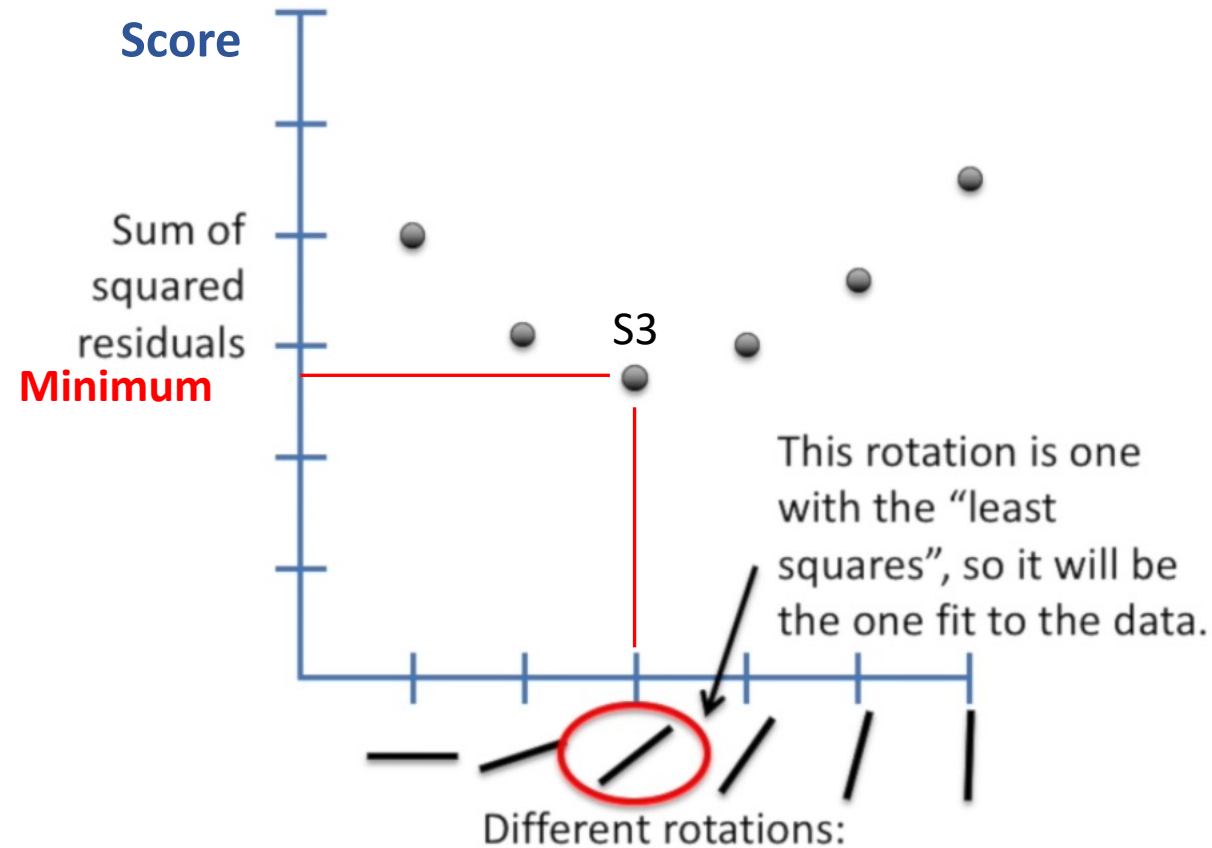
Least square method



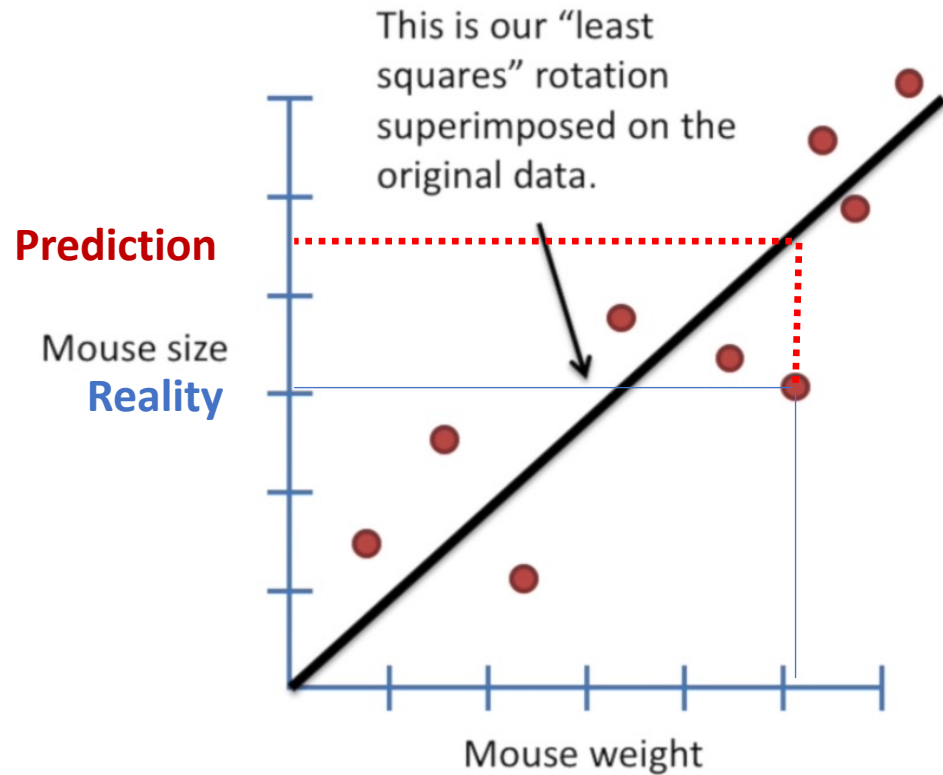


Again & again, recalculate

Resume : Sums of squared residuals for each rotation



Best rotation (=line position), the one which minimize the score of Sums of squared residuals !!!!



$$y = 0.1 + 0.78x$$

Dependence to « Mouse weight »

Coefficient R^2 = prediction quality

how good is the linear regression model to predict Mouse size taking into account Mouse weight!!

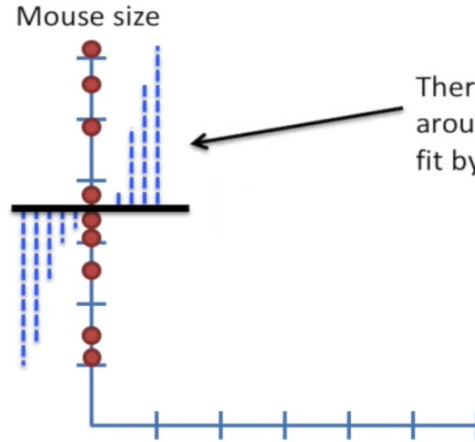
R^2 : Determination Coefficient

$$R^2 = \frac{\text{Var}(\text{mean}) - \text{Var}(\text{fit})}{\text{Var}(\text{mean})}$$

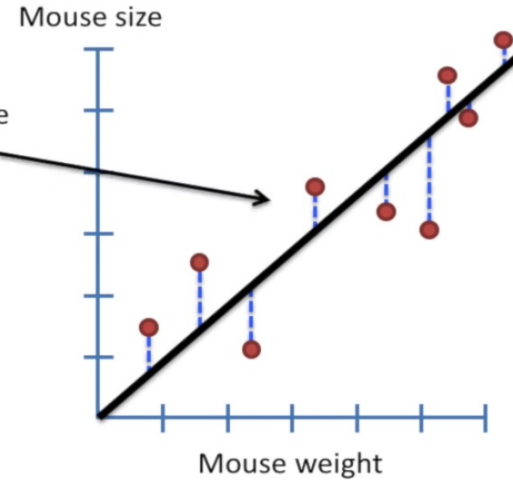
$$R^2 = \frac{\text{Variation expliquée}}{\text{Variation totale}}$$

$$\text{Var}(\text{mean}) = \frac{\text{SS}(\text{mean})}{n}$$

$$\text{Var}(\text{mean}) = 11.1$$



There is less variation around the line that we fit by least-squares.



$$\text{Var}(\text{fit}) = \frac{(\text{data} - \text{line})^2}{n}$$

$$\text{Var}(\text{fit}) = 4.4$$

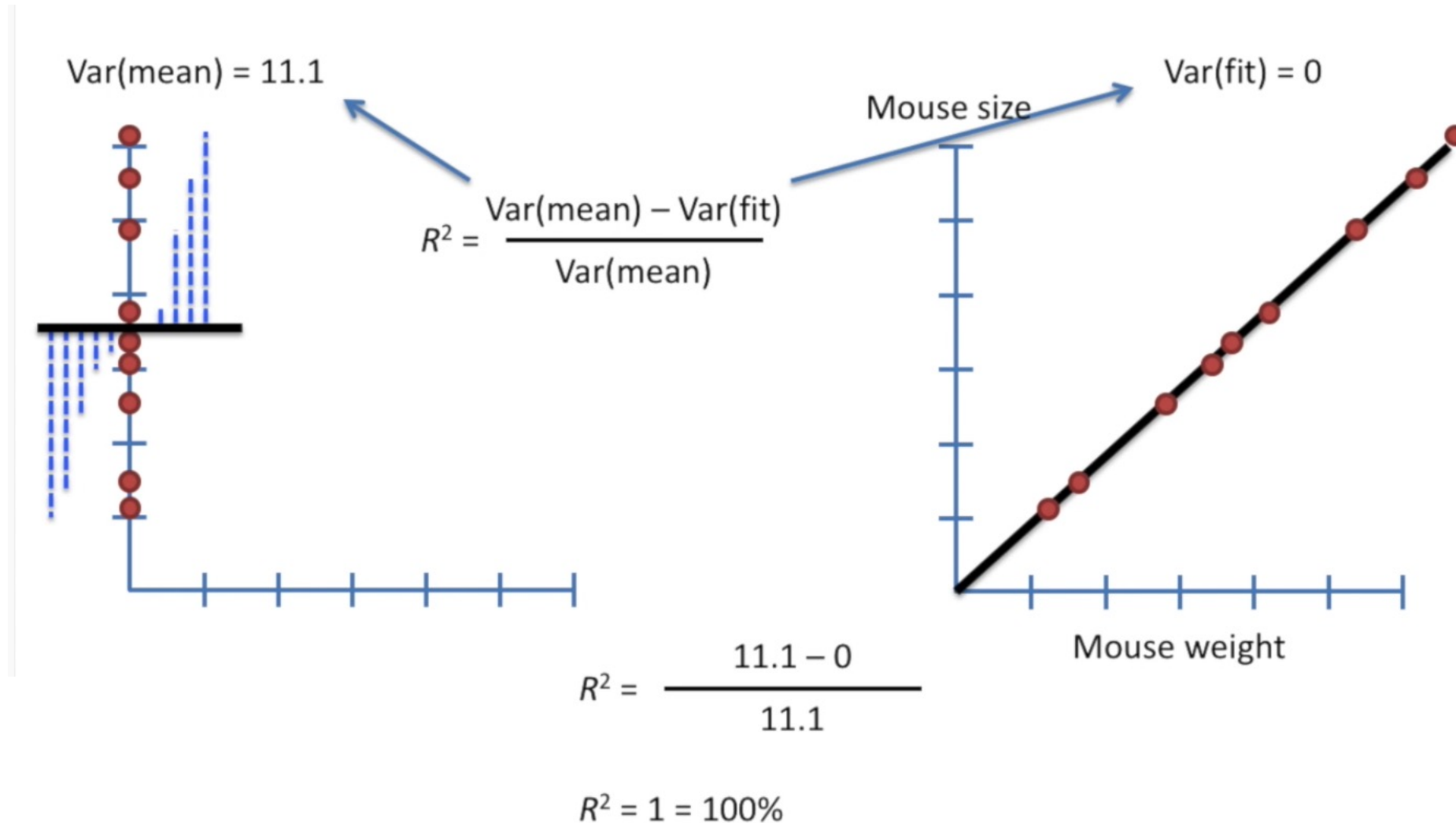
$$R^2 = \frac{11.1 - 4.4}{11.1} = 0.6 = 60\%$$

→ R^2 provides percentage variation in Y which is explained by all the X together = **% variation of the response variable explained by a linear model (weight variable)**

→ R^2 between 0 and 1

→ Here the established model explains 60% of the variability/variance of the "Mouse size »

TO be sure ...

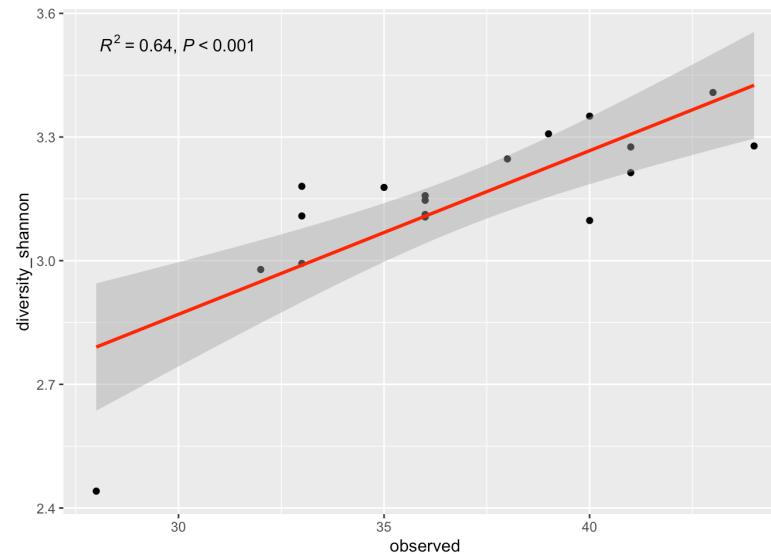


$R^2 = 1$ = max strength of linear regression model
(Lesser R^2 the value, the more scattered are the data points)

R^2 & significance?

R^2 based on variance comparison ... so p-value is given by the ratio F & distribution F

$$F = \frac{\text{The variation in mouse size explained by weight}}{\text{The variation in mouse size not explained by weight}}$$



Relation between r & R^2

Correlation coefficient of Pearson r can be linked to linear regression R^2
Its square is the explained variance by the regression (R^2)

$r = 0.5 \rightarrow R^2 = 0.25 \rightarrow 25\%$ of the Y variance explained by X variable... ☹️